

Overview of the Atlantic Sea Scallop Resource in the Gulf of Maine

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1. Introduction

The Atlantic sea scallop (*Placopecten magellanicus*) ranges from Cape Hatteras to the Gulf of St. Lawrence. The scallop fishery is primarily prosecuted in concentrated areas in and around Georges Bank and off the Mid-Atlantic coast, in waters extending from the near-coast out to the edge of the continental shelf. Atlantic sea scallops occur primarily in depths less than 110 meters on sand, gravel, shells, and cobble substrates (Hart and Chute 2004). While the majority of the Atlantic sea scallop resource is found on Georges Bank and in the Mid-Atlantic, sea scallops also occur in the Gulf of Maine (GOM) in both state and federal waters. The federal scallop resource in the GOM is managed by the New England Fishery Management Council (Council) and NOAA Fisheries, and has supported a growing directed fishery in recent years.

There have been several important changes to the state of scallop science and management in the GOM since the last Scallop benchmark assessment (2018) and management track assessment (2020). This working paper builds on earlier efforts to catalogue fishery dependent and independent data in the GOM region that were presented in SAW/SARC 59 and SAW/SAR 65. Over the long-term, the working group recommends that the scallop population in the GOM be included in stock assessment models and included in calculations for status determination.

Scallops in the GOM region are currently not considered in the Catch At Size Analysis (CASA; TOR 4) model and Stochastic Yield Model (SYM; TOR 5), and reference points for the area have not been developed. To develop catch advice for the directed fishery, the Scallop Areas Management Simulator (SAMS) projection model is used with recent year's survey data. When setting specifications for the overall fishery, scallops that are surveyed in the GOM (inclusive of the Northern Gulf of Maine [NGOM] management unit) are counted toward the legal limits (Overfishing Limit [OFL] and Acceptable Biological Catch [ABC]) for that fishing year.

In the absence of reference points and a stock assessment model for the GOM, the OFL and ABC estimates for the GOM are derived using the Georges Bank F_{MSY} estimates from the 2020 management track assessment ($F=0.46$ for OFL, $F=0.32$ for ABC). This approach was recommended by the NEFMC's SSC in October 2021.¹ Catch limits for the NGOM management area are set using target fishing mortality rates below the ABC proxy, and by regulation must be between $F=0.15$ and $F=0.25$.

2. Gulf of Maine Region

The Gulf of Maine is characterized by a series of banks and ledges, along with deep basins (Figure 1). Scallops are generally found in sand, cobble, and pebble habitats on these banks and ledges, though they are known to occur in less-suitable mud-silt areas in deeper water (Figure 3). Within the GOM region, the NGOM Management Area covers the majority of inshore areas, starting at $42^{\circ} 20' N$ (roughly Boston, MA) and extending north to the Canadian border (Figure 2). The southern boundary of this management unit bisects Stellwagen Bank, which is a productive scallop fishing ground.

¹ Additional information at: <https://www.nefmc.org/library/december-2021-ssc>

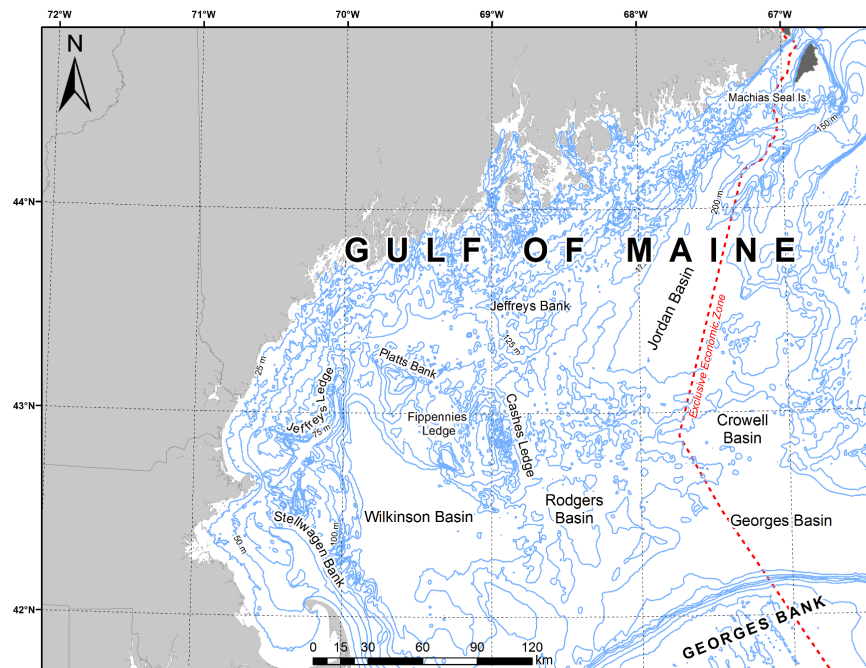


Figure 1 - Major physiographic features of the Gulf of Maine relative to the territorial waters boundary of the US and Canada (EEZ, red dotted line).

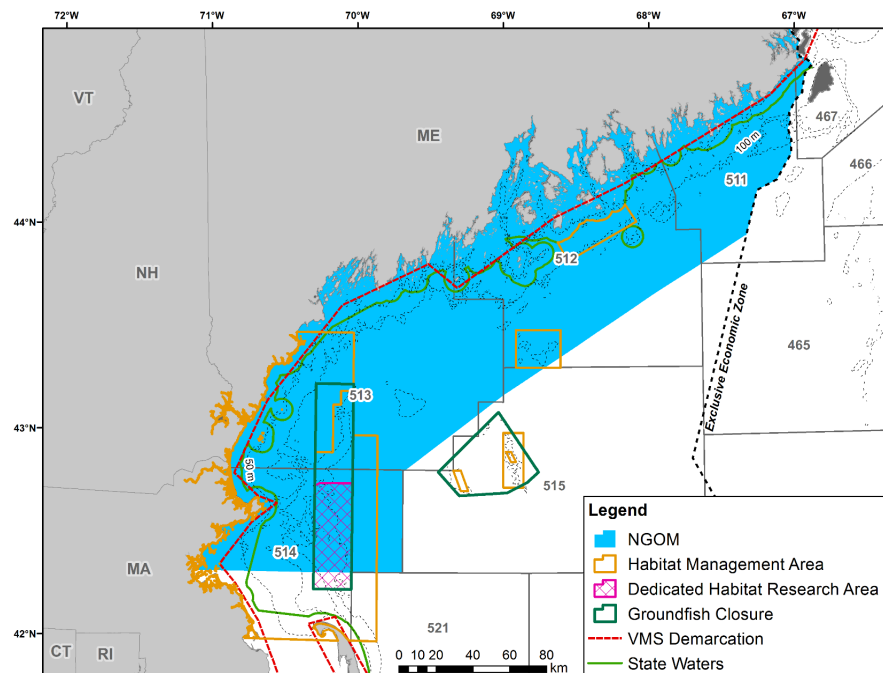


Figure 2 - Management boundaries in the Gulf of Maine region including boundaries of the Northern Gulf of Maine Management Area (NGOM, in blue), Habitat Management Areas, Dedicated Habitat Research Areas, Groundfish Closures, VMS Demarcation line, State Waters, and Statistical Reporting Areas (grey lines and labels).

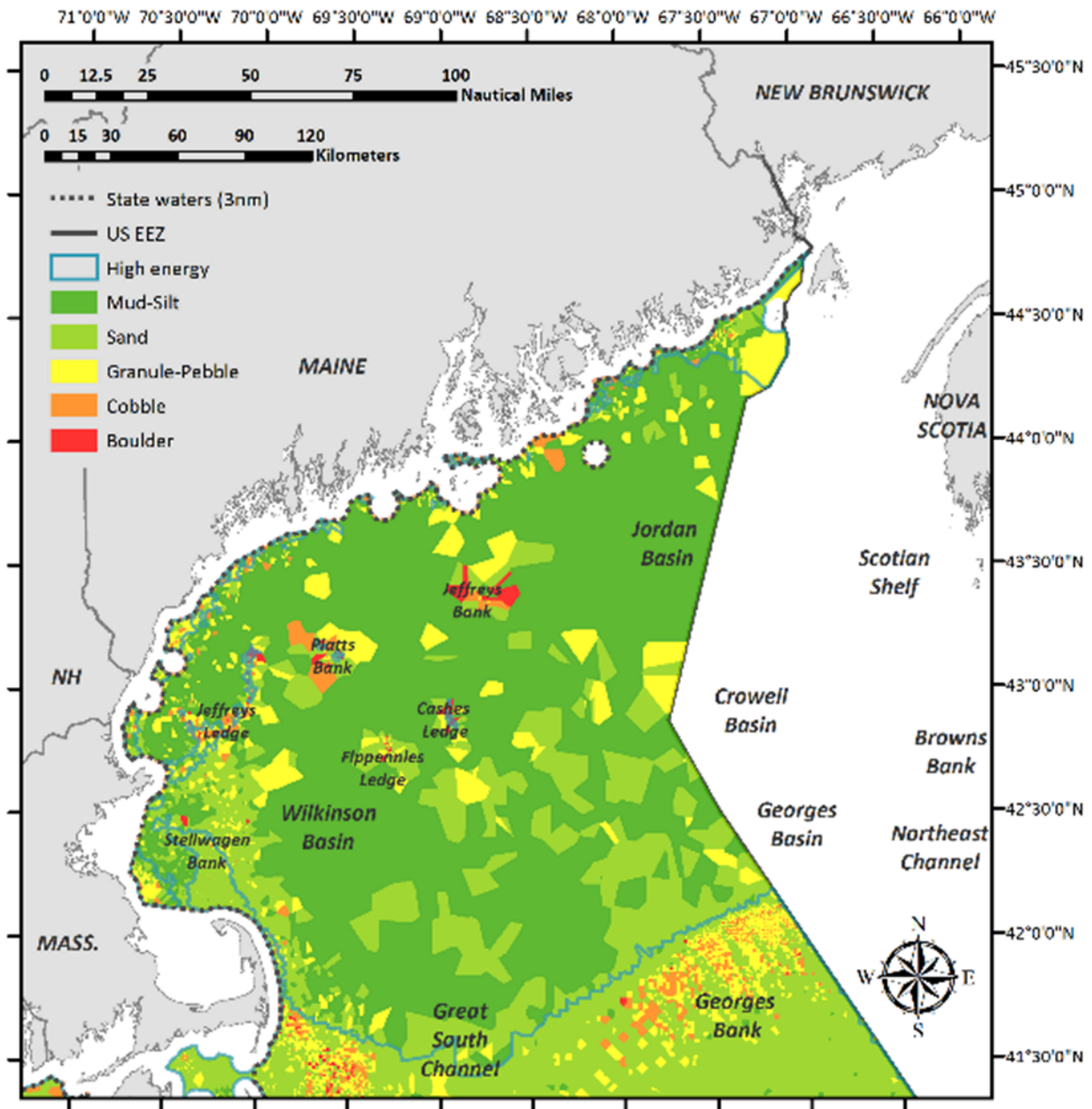


Figure 3 - Sedimentary features of the Gulf of Maine (NEFMC, 2016).

3. Fishery Independent Surveys

Fishery independent surveys in the GOM were periodic in the early 1980's, but have become more frequent in recent years through the support of the Research Set-Aside (RSA) program and industry partners. Truesdell et al (2014) noted in SAW59 that there

was a limited amount of fishery independent data available for the GOM Region (Figure 1). Dredge and optical surveys have occurred more regularly since the last benchmark assessment, however the region remains data-limited relative to Georges Bank and the Mid-Atlantic. Some of the earliest dedicated scallop surveys of the region took place in the early 1980's (Serchuk, 1983). Recent fishery independent surveys have been conducted by the School for Marine Science and Technology (SMAST) and Maine Department of Marine Resources (DMR).

School for Marine Science and Technology (SMAST) Survey Overview

Following the methodology described in TOR 3, SMAST students and scientists have conducted drop camera surveys on various banks and ledges in the GOM almost annually since 2009 (Figure 4). The areas surveyed are banks and ledges in the GOM, but with some split into sections to reflect management boundaries (Figure 5). Stellwagen Bank is located in the south of the GOM and is split into three regions in the SMAST figures. Samples from Stellwagen Bank that fall within the Western Gulf of Maine (WGOM) closure area are referred to as "WGOM". The remaining Stellwagen Bank samples are separated further by the NGOM management boundary, where samples outside of this area in Stellwagen Bank are referred to as "Stellwagen South", and those within the boundary are "Stellwagen NGOM". Jeffreys Ledge further north is split into two areas by the WGOM closure area boundary, with samples inside this area referred to as "Jeffreys WGOM" and those outside the boundary called "Jeffreys". To the west of Jeffreys, Ipswich Bay is split into two groups based on the boundary of Massachusetts state waters. Samples from within state waters are referred to as "Ipswich State", and those falling in federal waters are called "Ipswich Federal". The three other areas surveyed by SMAST were Platts Bank, Cashes Ledge, and Fippennies Ledge, which are not divided any further.

The GOM surveys commonly used a fine-scale systematic grid size, with stations placed 1 km apart to accommodate the relatively small, non-contiguous nature of the GOM banks and ledges. Some coarser grid resolutions were also used in some years. Similar to the Georges Bank and Mid-Atlantic sampling, the GOM surveys were conducted using an SD video camera up until 2017, and from 2017 onwards the surveys used a high-resolution digital still camera. Mean scallop densities for a range of size classes at each station were calculated and shell height frequencies were collated by area (Figures 6 to 19).

SMAST has also operated a dredge survey annually since 2022 that collected data to develop shell height to meat weight relationships to convert drop camera scallop counts and shell height measurements to biomass estimates in the GOM. The scientific dredge used was an eight foot-wide New Bedford-style dredge with two inch rings. The 2022

survey focussed on a small number of tows (six) that were conducted during the drop camera survey and in areas deemed to have sufficiently high scallop densities from the images and video taken by the drop camera (Figure 20). Each tow lasted approximately twenty minutes and aimed to catch 150 scallops. Tows were repeated in the same area until 150 scallops suitable for dissections were caught before then moving on to look for other suitable areas with high densities of scallops on Stellwagen Bank, Jeffreys Ledge, and in Ipswich Bay.

The 2023 survey used a different sampling approach, where thirty-six five-minute-long dredge tows were conducted throughout the same three areas. As in 2022, station locations were targeted towards high density locations from the 2022 drop camera survey observations. Only fifteen scallops were targeted for dissections from each tow for the 2023 survey, and repeat tows over the same target station were not conducted. The change in approach from the 2022 survey was to increase the survey coverage within the GOM, whilst maintaining a similar number of scallops to be dissected. Neither survey conducted tows within the WGOM closure area. A new dredge with the same specifications as the 2022 survey (eight foot-wide New Bedford-style dredge with two inch rings, rock chains, a 1 $\frac{3}{4}$ " headbail, and 4 $\frac{1}{2}$ " twine top) was used. After each tow, the shell height of each scallop was measured to the nearest millimeter and the adductor muscle, hereafter meat, was dissected and weighed. In the 2023 survey the gonad weight and remaining soft tissue weight were also measured to the nearest millimeter. The resultant data were combined with shell height and meat weight samples obtained from the same three areas by the Maine Department of Marine Resources in 2016, 2019, and 2021 surveys described in this Appendix. The combined shell height and meat weight dataset was used to fit a shell height to meat weight relationship that enabled the conversion of shell height measurements and count data from the drop camera survey to biomass in 2023. The shell height meat weight relationship was fit as a Gamma family linear mixed effects model with natural logarithm shell height, natural logarithm depth, and latitude as fixed covariates and tow number and bank as random covariates, which is consistent with the methods used for Georges Bank and the Mid Atlantic from SARC 65. Previous years implemented other shell height to meat weight relationships (Table 1) derived from other sampling efforts on Georges Bank to convert drop camera abundance to biomass estimates.

The gonad weight data was used to fit a shell height gonad weight relationship using only SMAST data from the 2023 survey, which was the only survey that gathered these data (Figure 21). The relationship was fit using identical model structure to the shell height meat weight equation developed from the SMAST and Maine DMR surveys in 2023. The gonad weights were also used to calculate an individual gonad somatic index for each scallop gathered during this survey (Figure 22). The gonad somatic index

was calculated as the weight of the gonad divided by the combined weight of the abductor muscle, gonad, and all remaining soft tissue. Shell weight was not measured and not incorporated into this calculation.

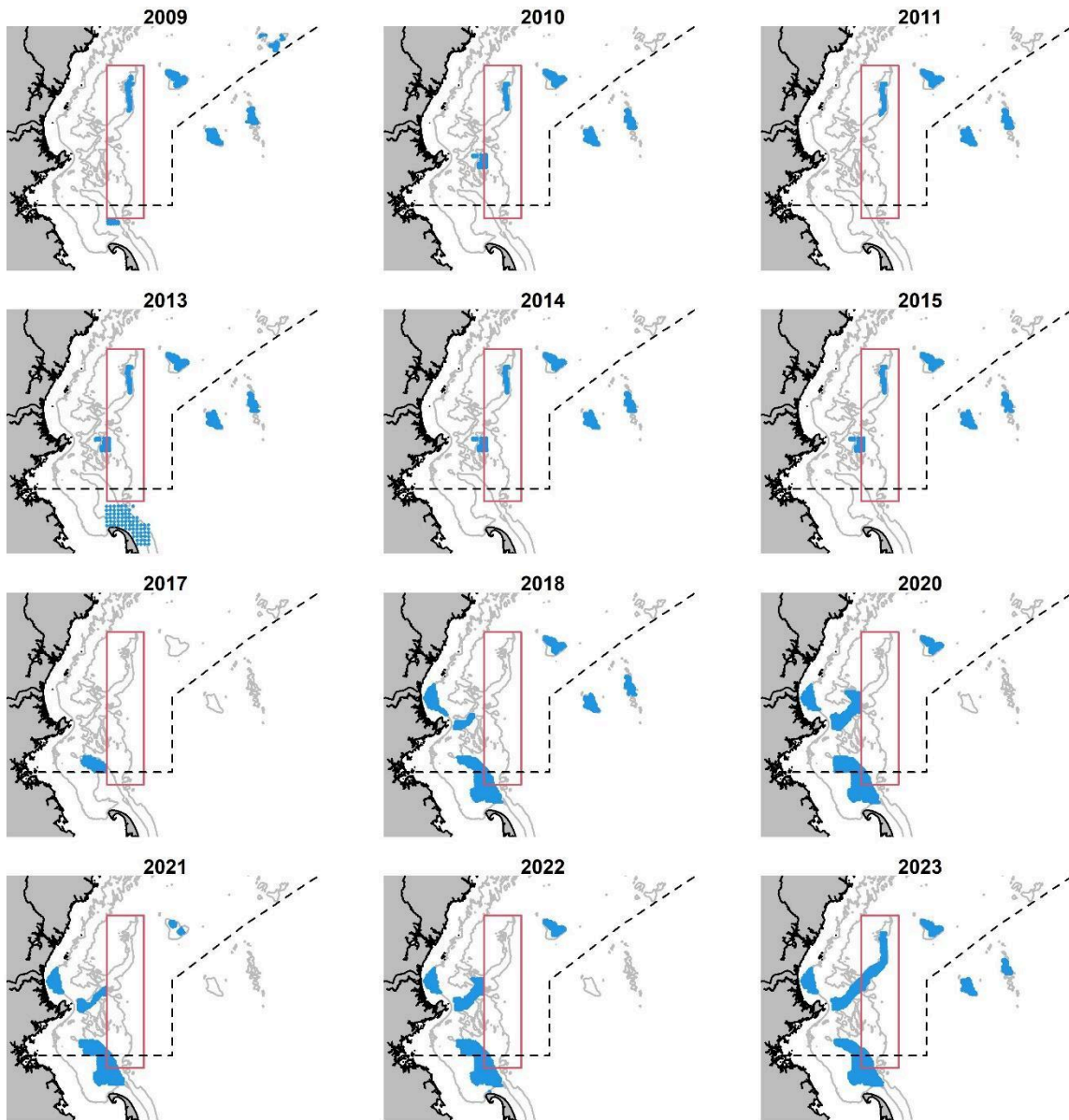


Figure 4. SMAST drop camera survey coverage (solid blue) in the Gulf of Maine by year. The red box surrounds the Western Gulf of Maine closure area, and the dashed black line denotes the southern and eastern boundaries of the Northern Gulf of Maine management area.

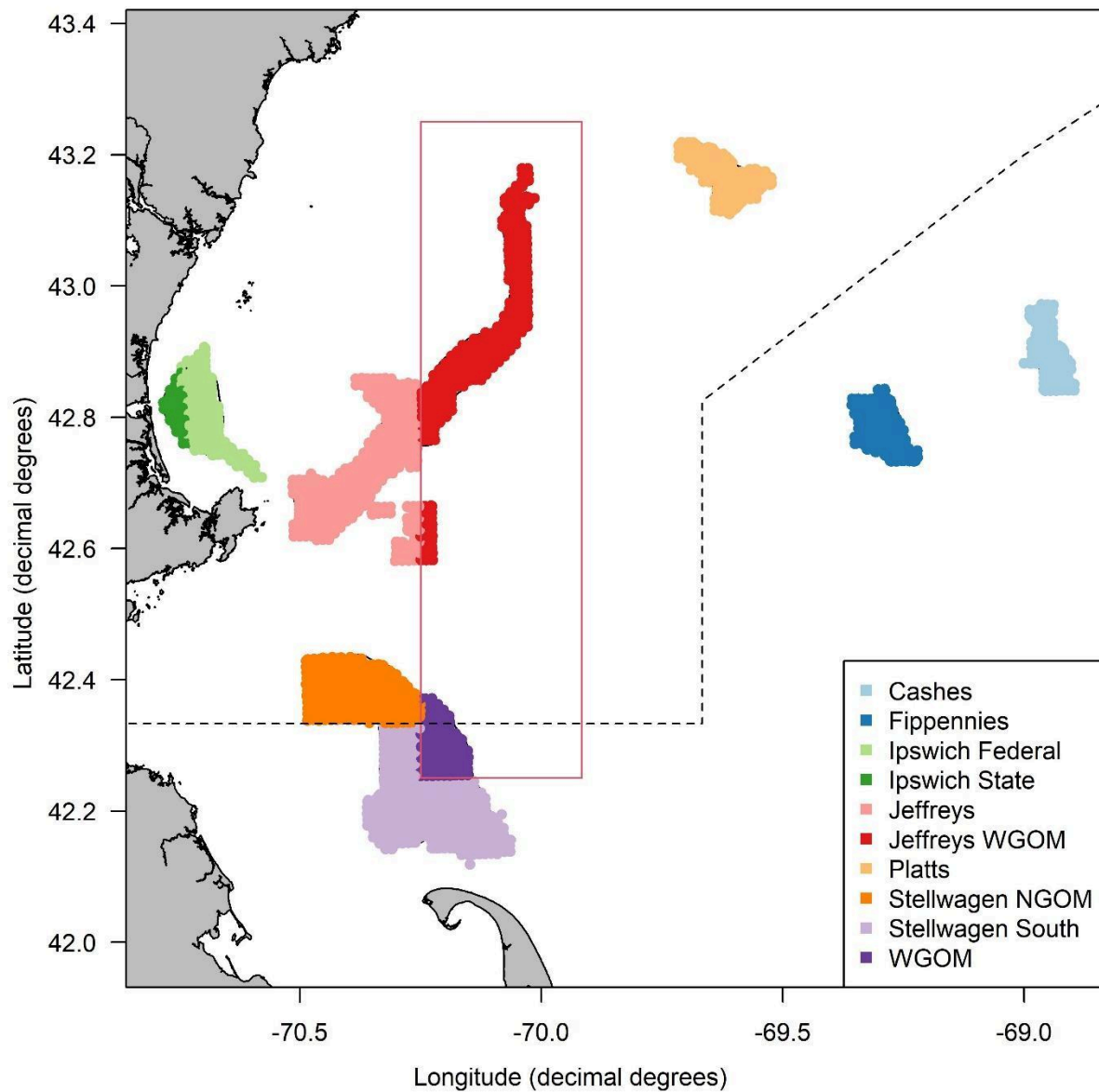


Figure 5. The areas and names of the Gulf of Maine where SMAST estimated scallop abundance and biomass. The red box surrounds the Western Gulf of Maine closure area and the dashed black line denotes the southern and eastern boundaries of the Northern Gulf of Maine management area.

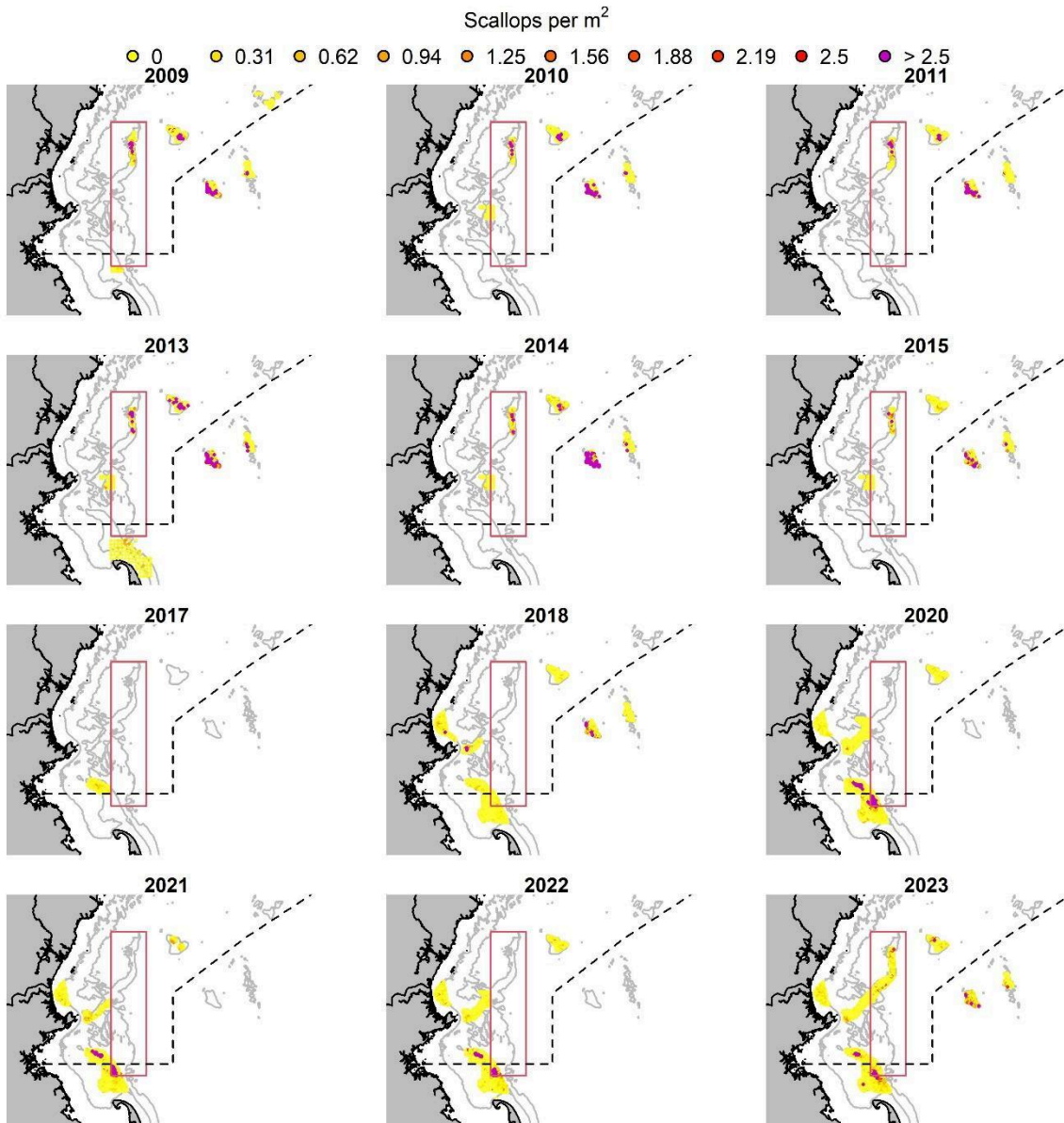


Figure 6. Scallop density (count per m^2) heatmap from the SMAST drop camera survey in the Gulf of Maine by year. All sizes of scallops measured are represented. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and therefore, surveys since 2017 are more likely to detect smaller scallops. The red box is the Western Gulf of Maine closure area, and the dashed black line is the southern and eastern boundaries of the Northern Gulf of Maine management area.

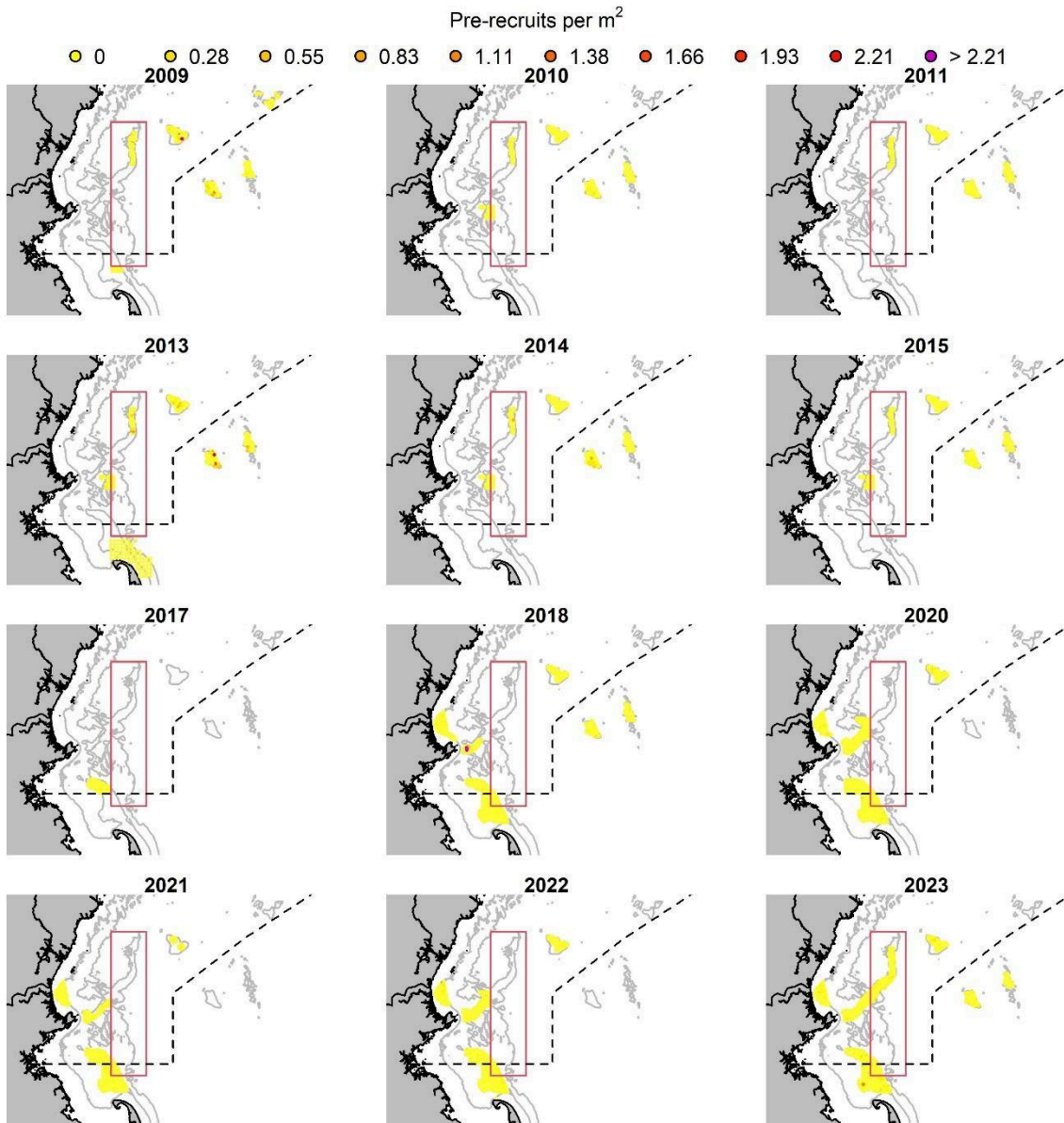


Figure 7. Scallop density (count per m²) heatmap from the SMAST drop camera survey in the Gulf of Maine by year showing pre-recruit scallops (less than 35 mm shell height) measured. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and therefore, surveys since 2017 are more likely to detect smaller scallops. The red box is the Western Gulf of Maine closure area, and the dashed black line is the southern and eastern boundaries of the Northern Gulf of Maine management area.

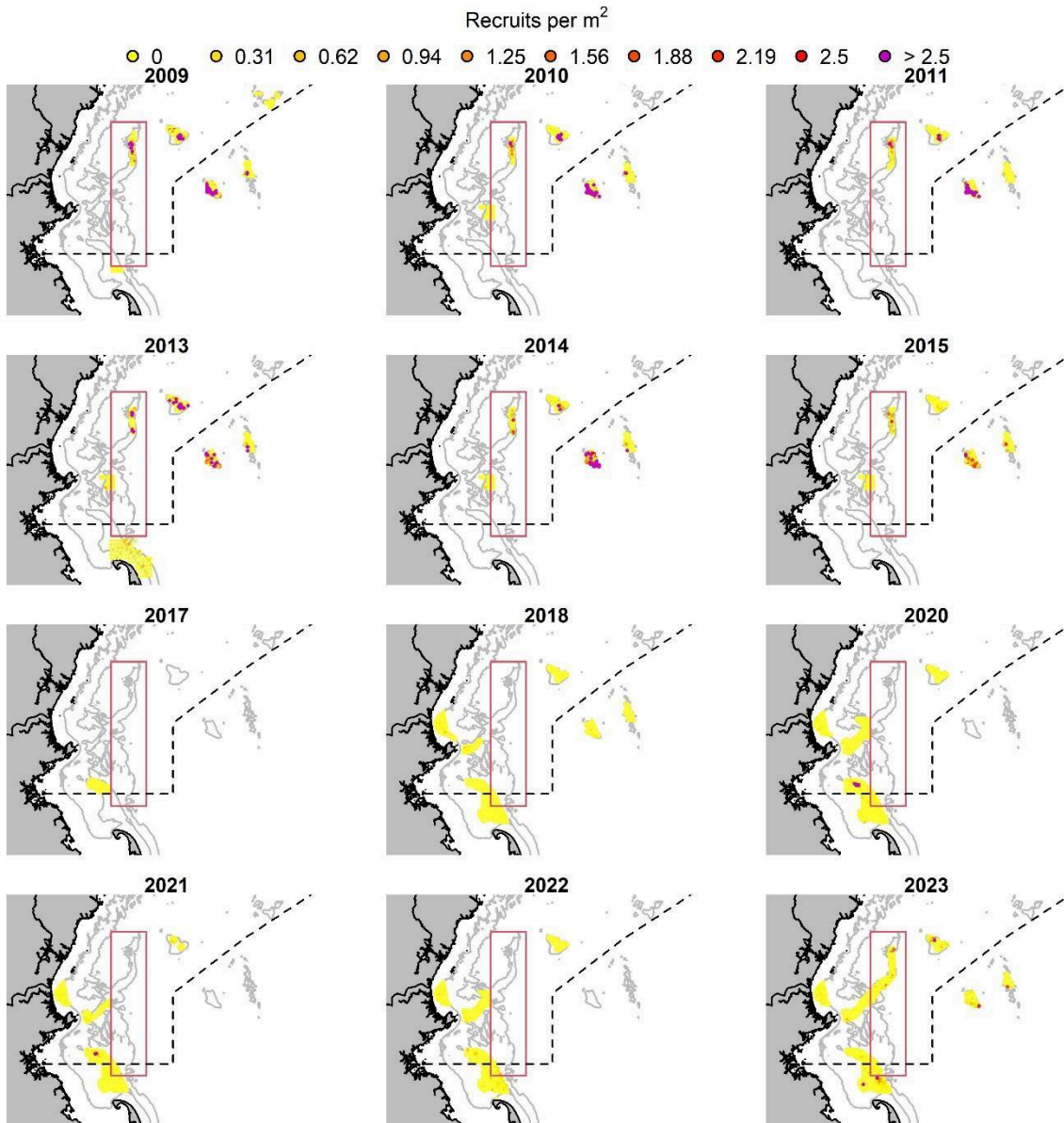


Figure 8. Scallop density (count per m²) heatmap from the SMAST drop camera survey in the Gulf of Maine by year, showing recruit scallops (between 35 mm and 75 mm shell height) measured. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and therefore, surveys since 2017 are more likely to detect smaller scallops. The red box is the Western Gulf of Maine closure area, and the dashed black line is the southern and eastern boundaries of the Northern Gulf of Maine management area.

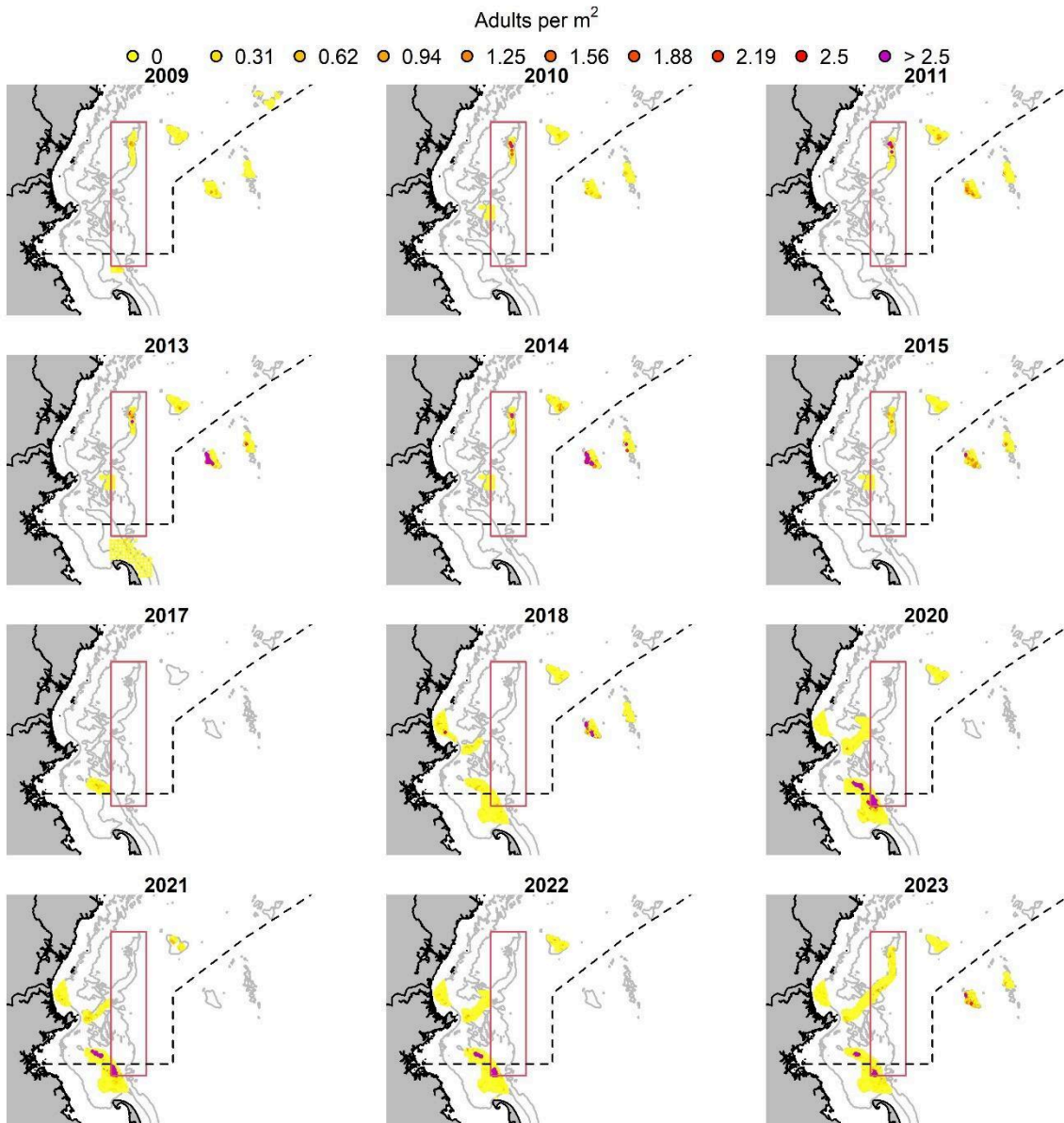


Figure 9. Scallop density (count per m²) heatmap from the SMAST drop camera survey in the Gulf of Maine by year, showing adult scallops (greater than 75 mm shell height) measured. The red box is the Western Gulf of Maine closure area, and the dashed black line is the southern and eastern boundaries of the Northern Gulf of Maine management area.

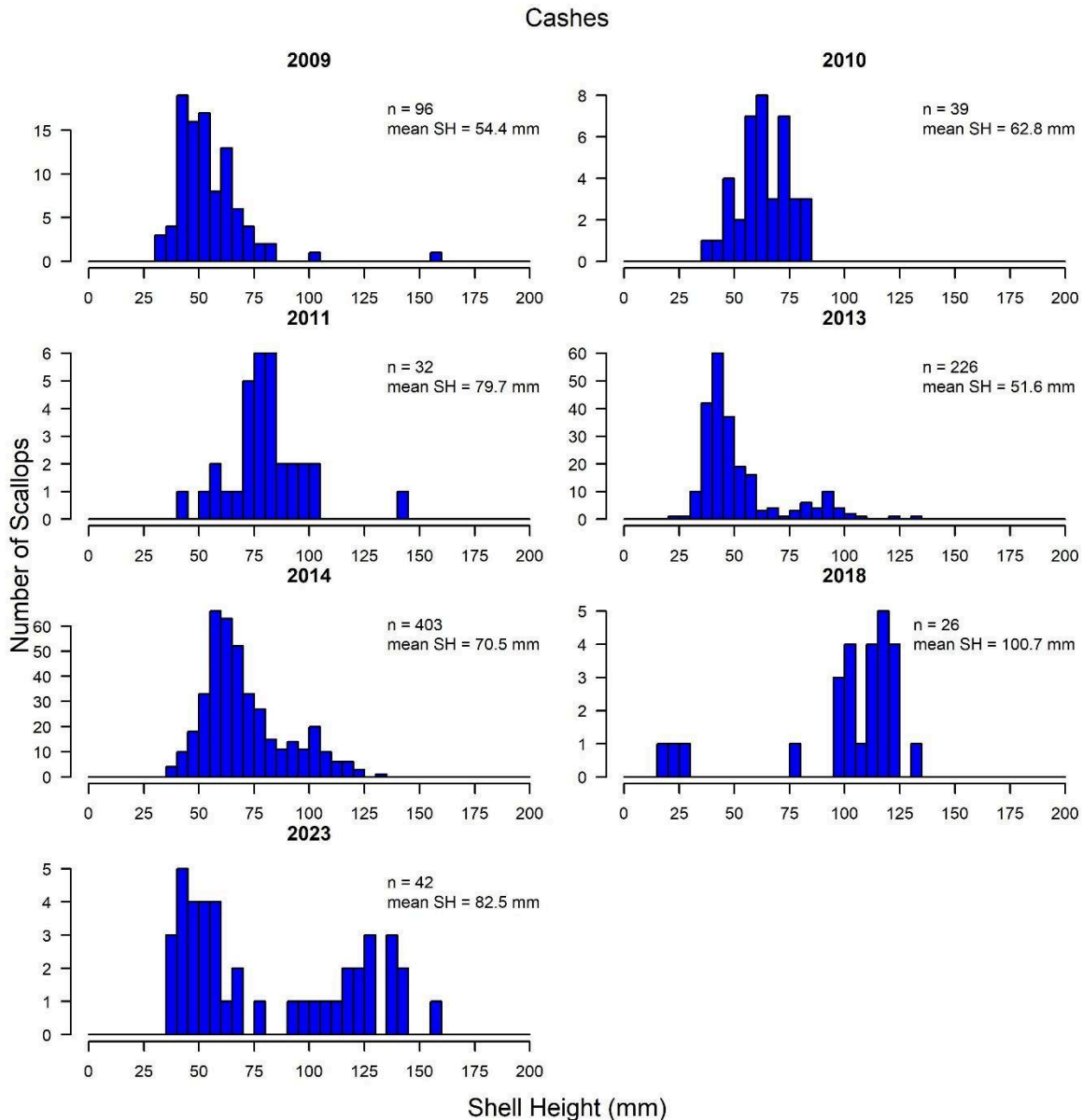


Figure 10. Scallop shell height frequencies from the SMAST drop camera survey on Cashes Ledge by year. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

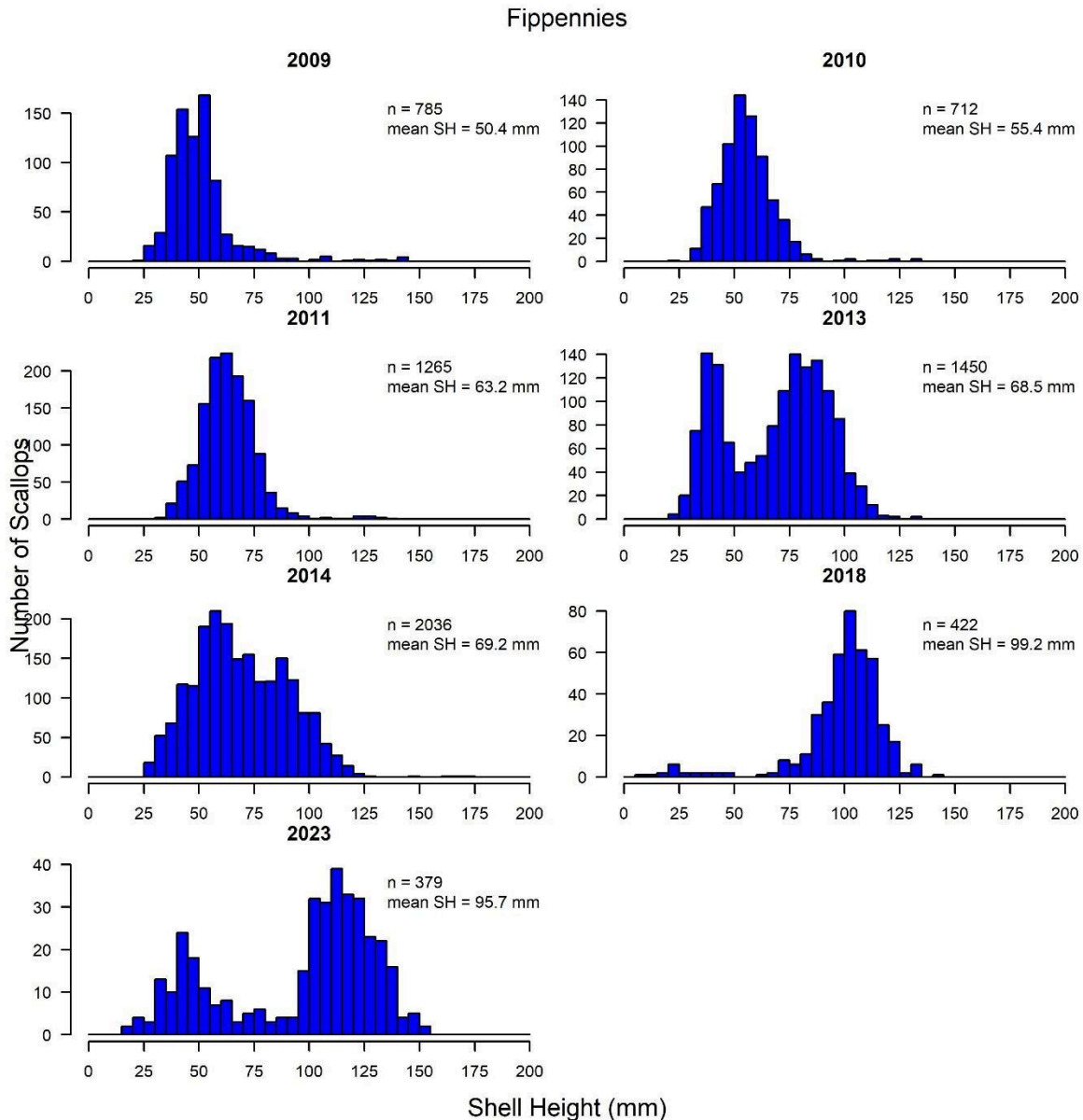


Figure 11. Scallop shell height frequencies from the SMAST drop camera survey on Fippennies Ledge by year. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

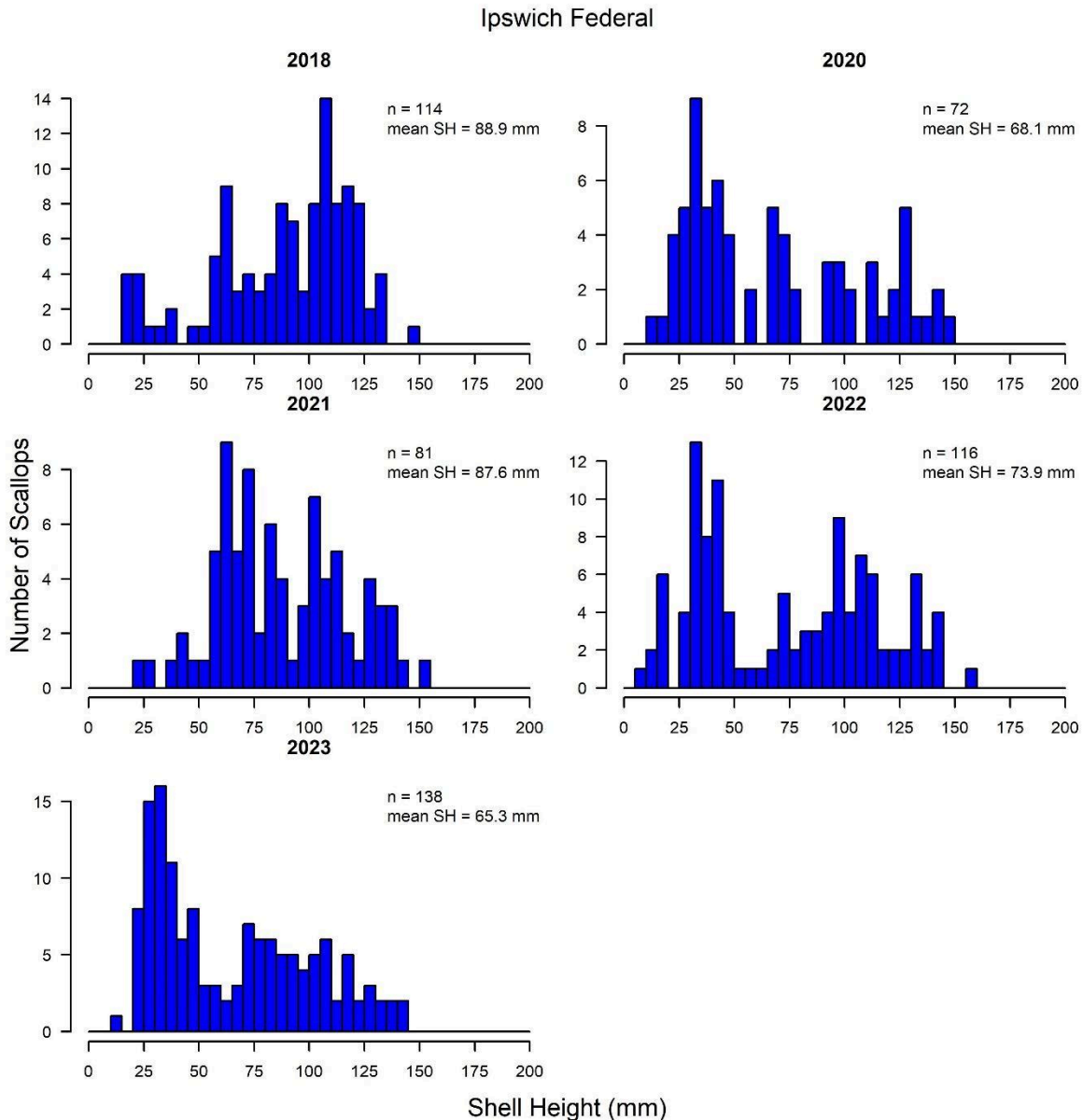


Figure 12. Scallop shell height frequencies from the SMAST drop camera survey of the part of Ipswich Bay that falls within federal waters. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

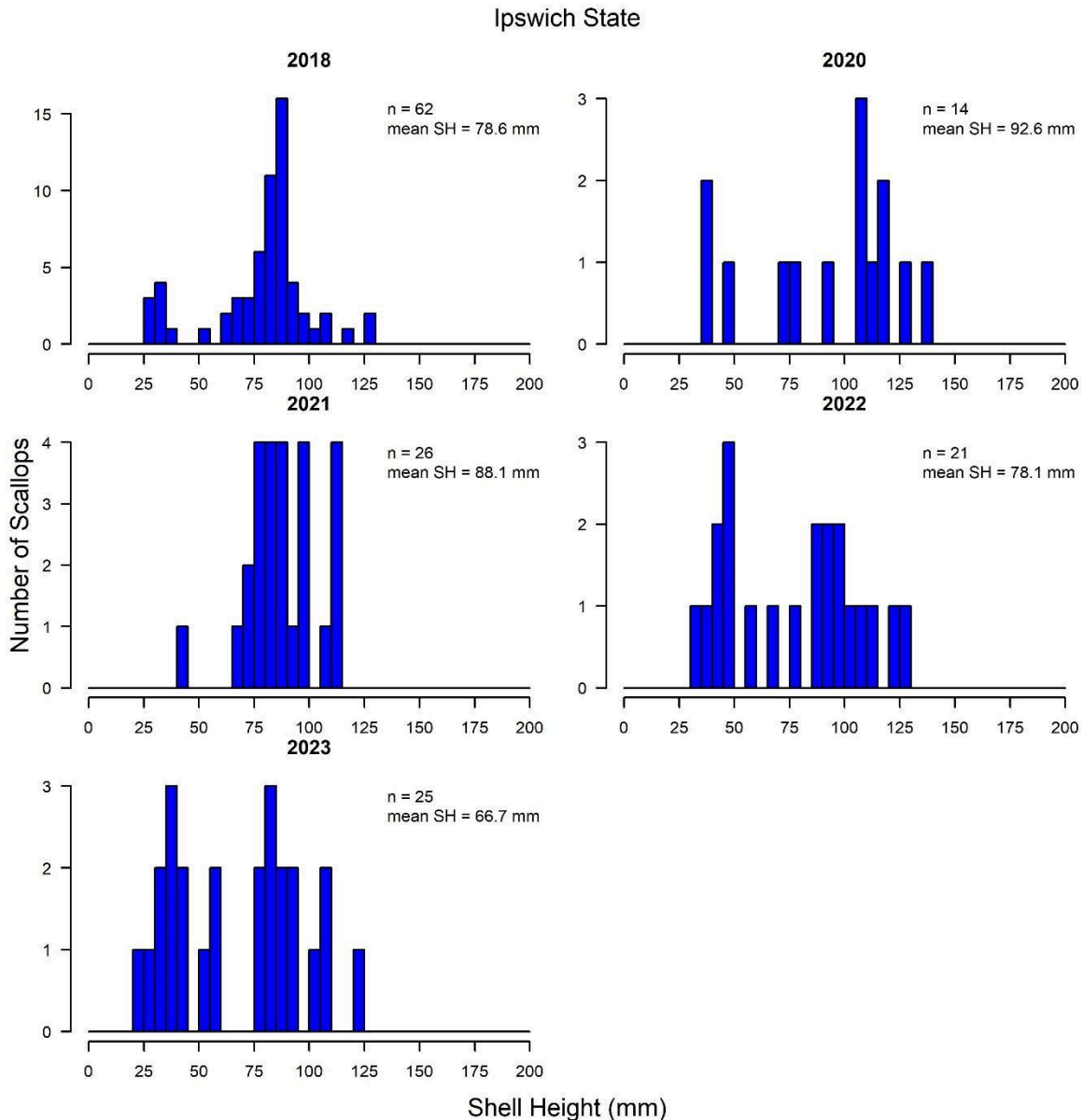


Figure 13. Scallop shell height frequencies from the SMAST drop camera survey of the part of Ipswich Bay that falls inside Massachusetts state waters. The total number of scallops measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

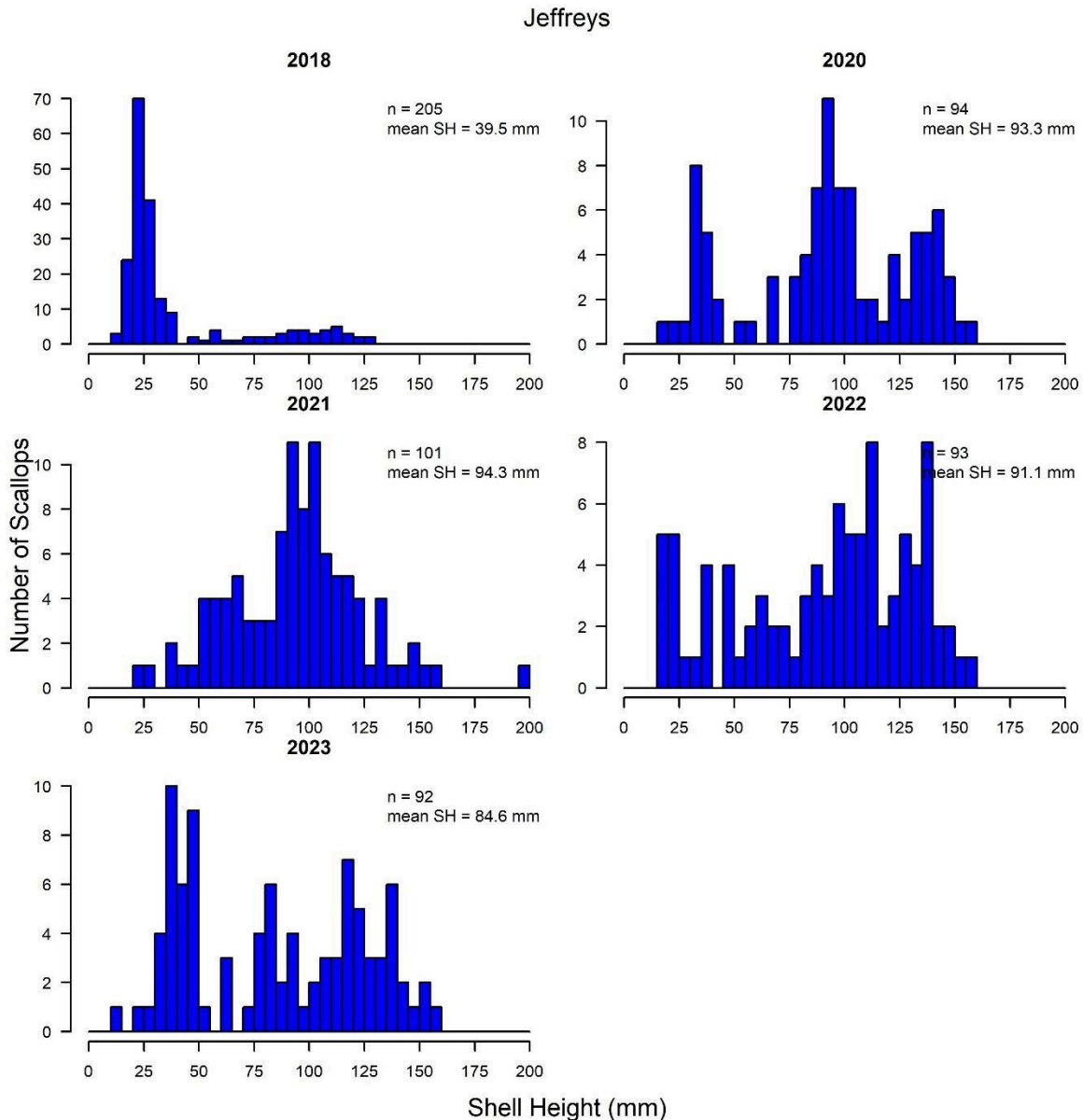


Figure 14. Scallop shell height frequencies from the SMAST drop camera survey of the part of Jeffreys Ledge outside of the Western Gulf of Maine closure area. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

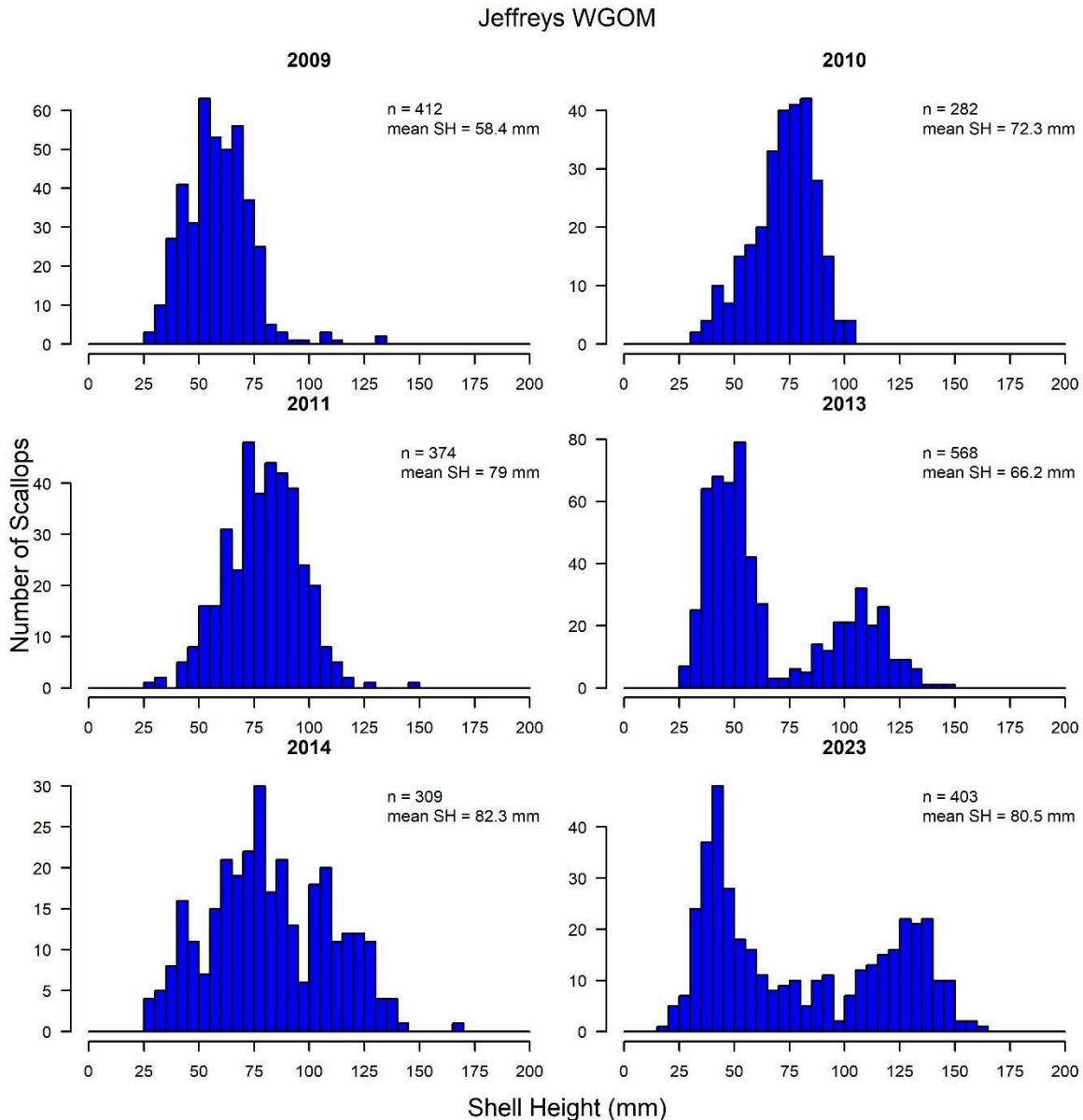


Figure 15. Scallop shell height frequencies from the SMAST drop camera survey of the part of Jeffreys Ledge inside the Western Gulf of Maine closure area. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

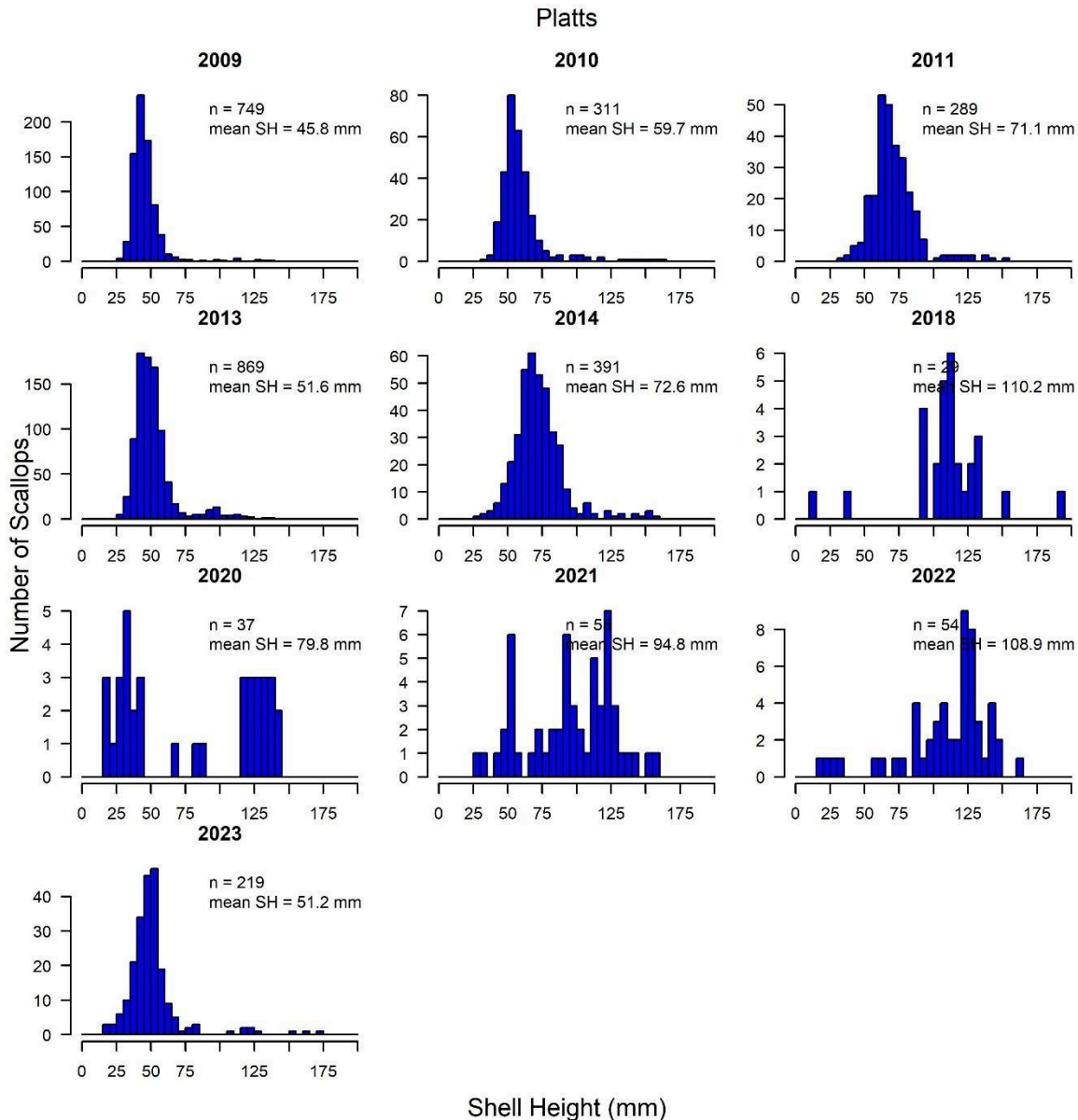


Figure 16. Scallop shell height frequencies from the SMAST drop camera survey on Platts Bank. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

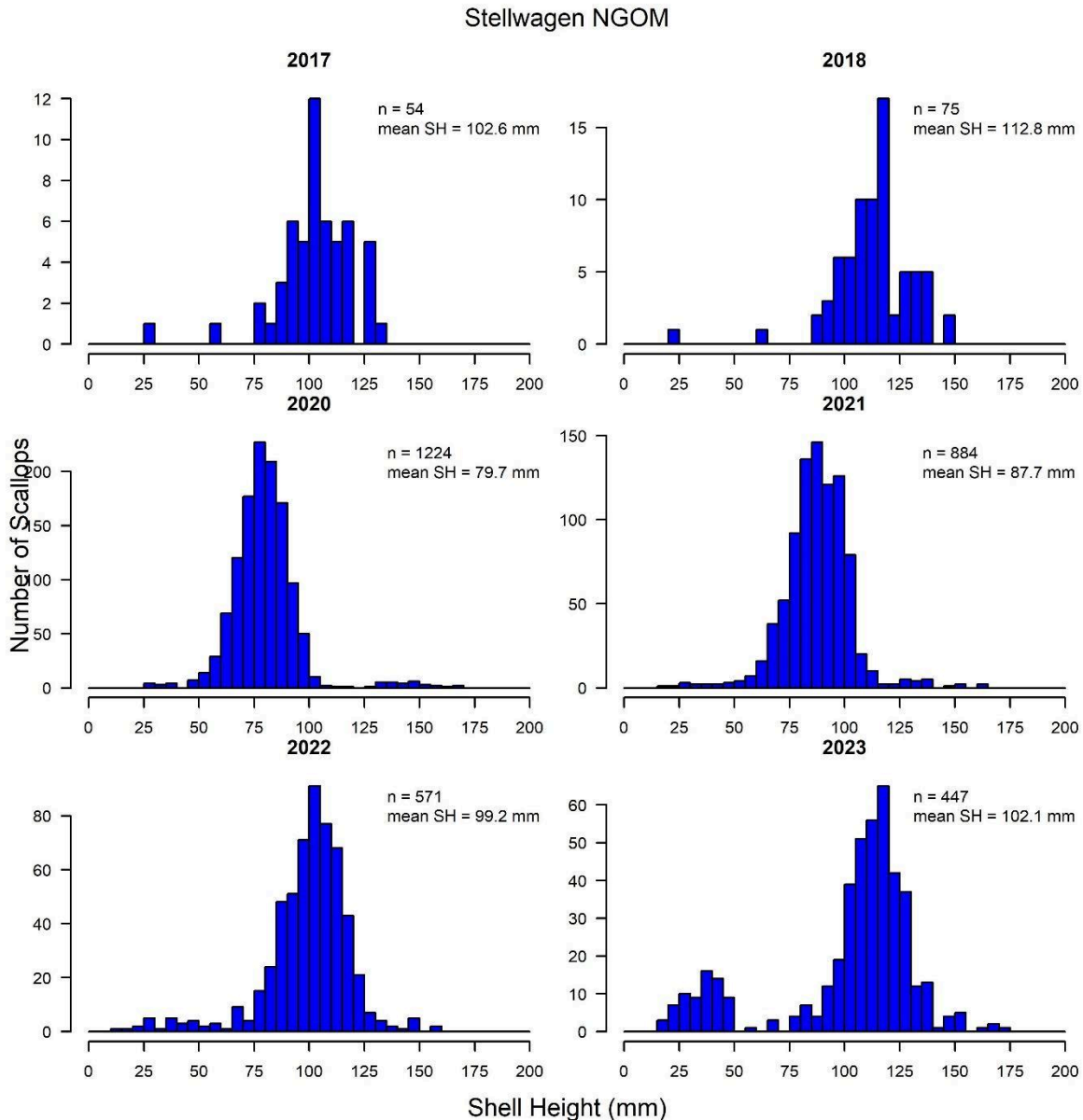


Figure 17. Scallop shell height frequencies from the SMAST drop camera survey on the part of Stellwagen Bank that is within the Northern Gulf of Maine management area, but outside the Western Gulf of Maine closure area. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

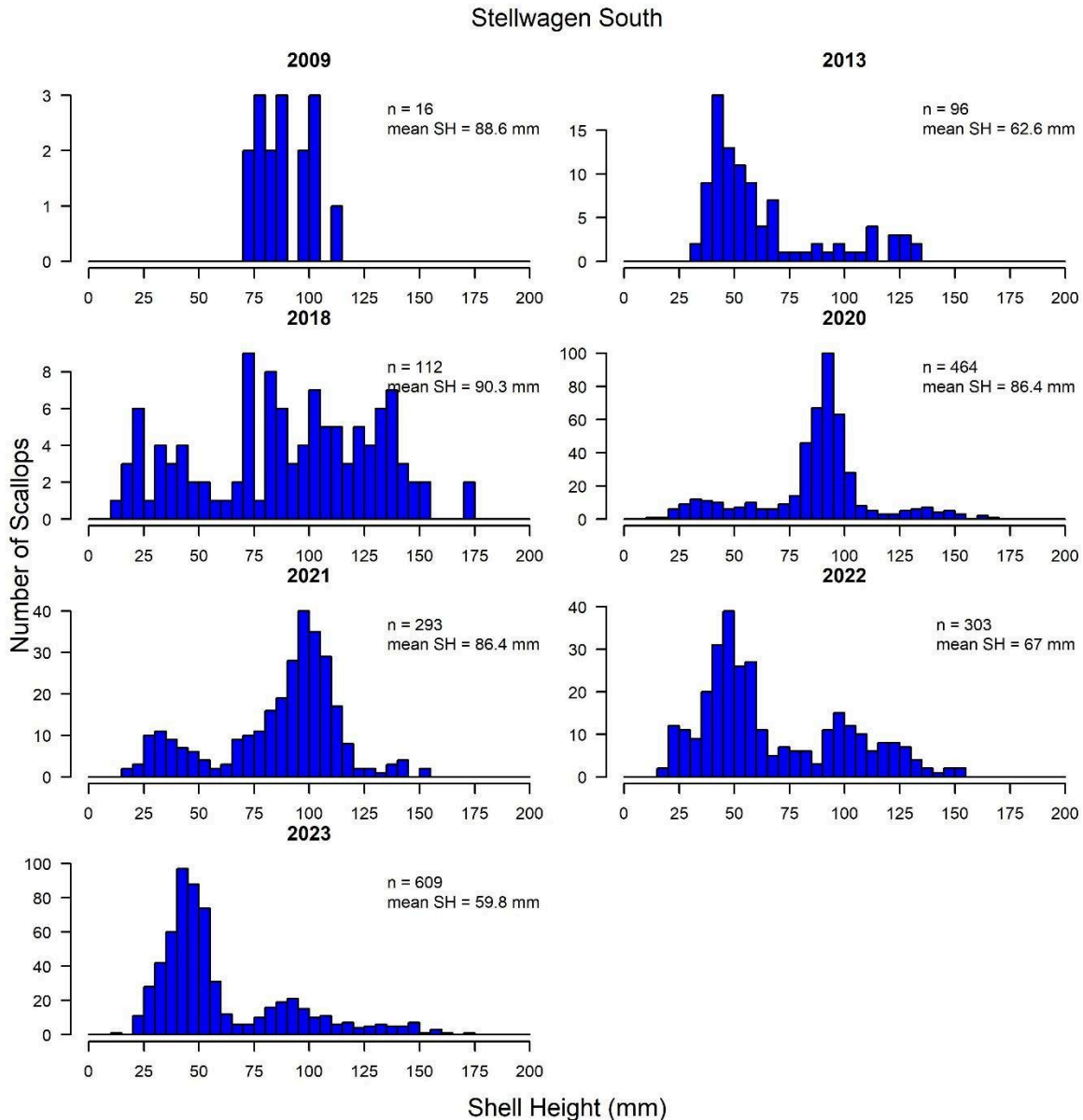


Figure 18. Scallop shell height frequencies from the SMAST drop camera survey on the part of Stellwagen Bank that is south of the Northern Gulf of Maine management area and outside of the Western Gulf of Maine closure area. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

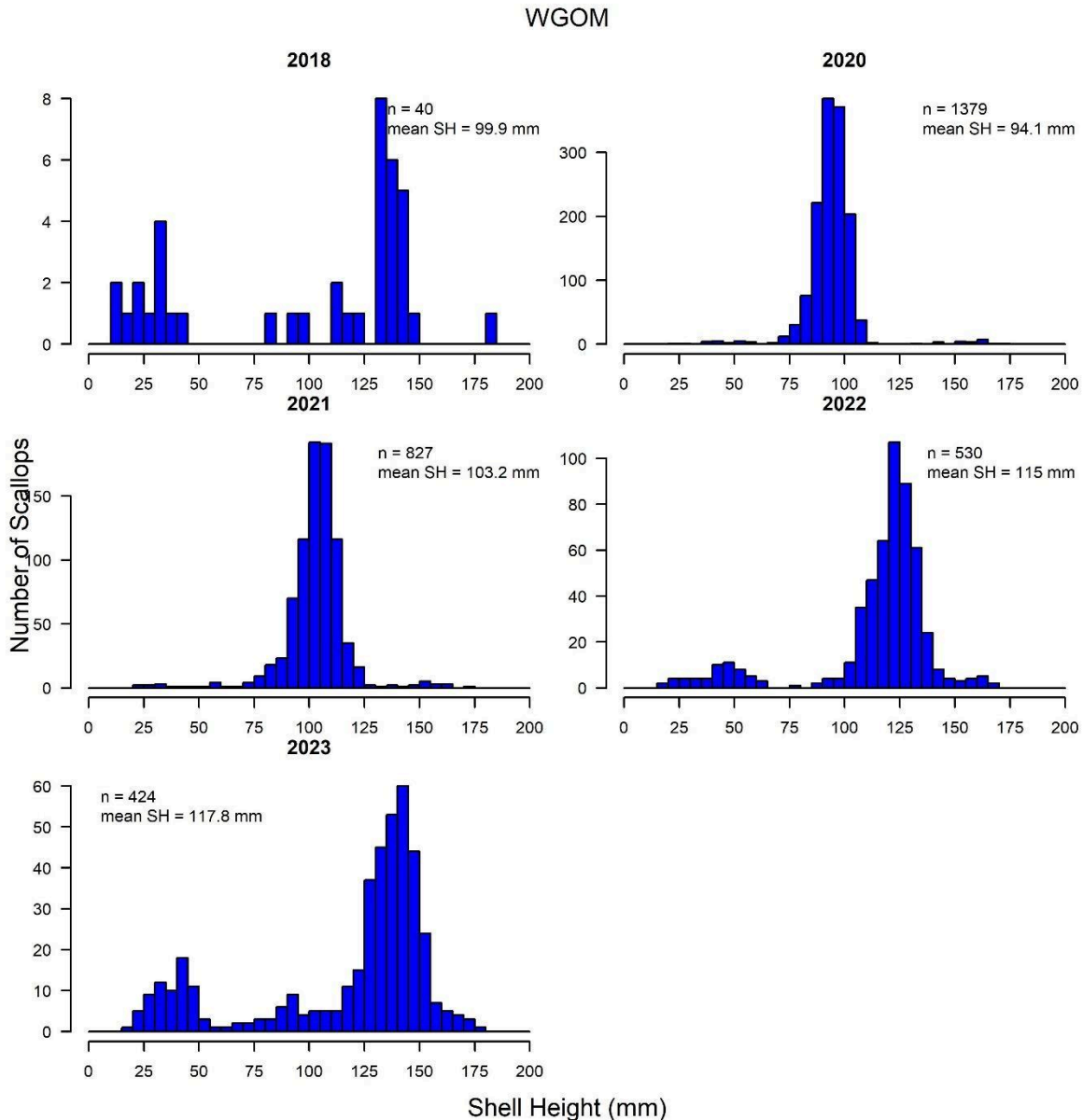


Figure 19. Scallop shell height frequencies from the SMAST drop camera survey on the part of Stellwagen Bank that is within the Western Gulf of Maine closure area. The total number measured and mean shell height (mm) by year are also displayed. Note that prior to 2017 a camera with a lower resolution was used as the main source of data and, therefore, surveys since 2017 are more likely to detect smaller scallops.

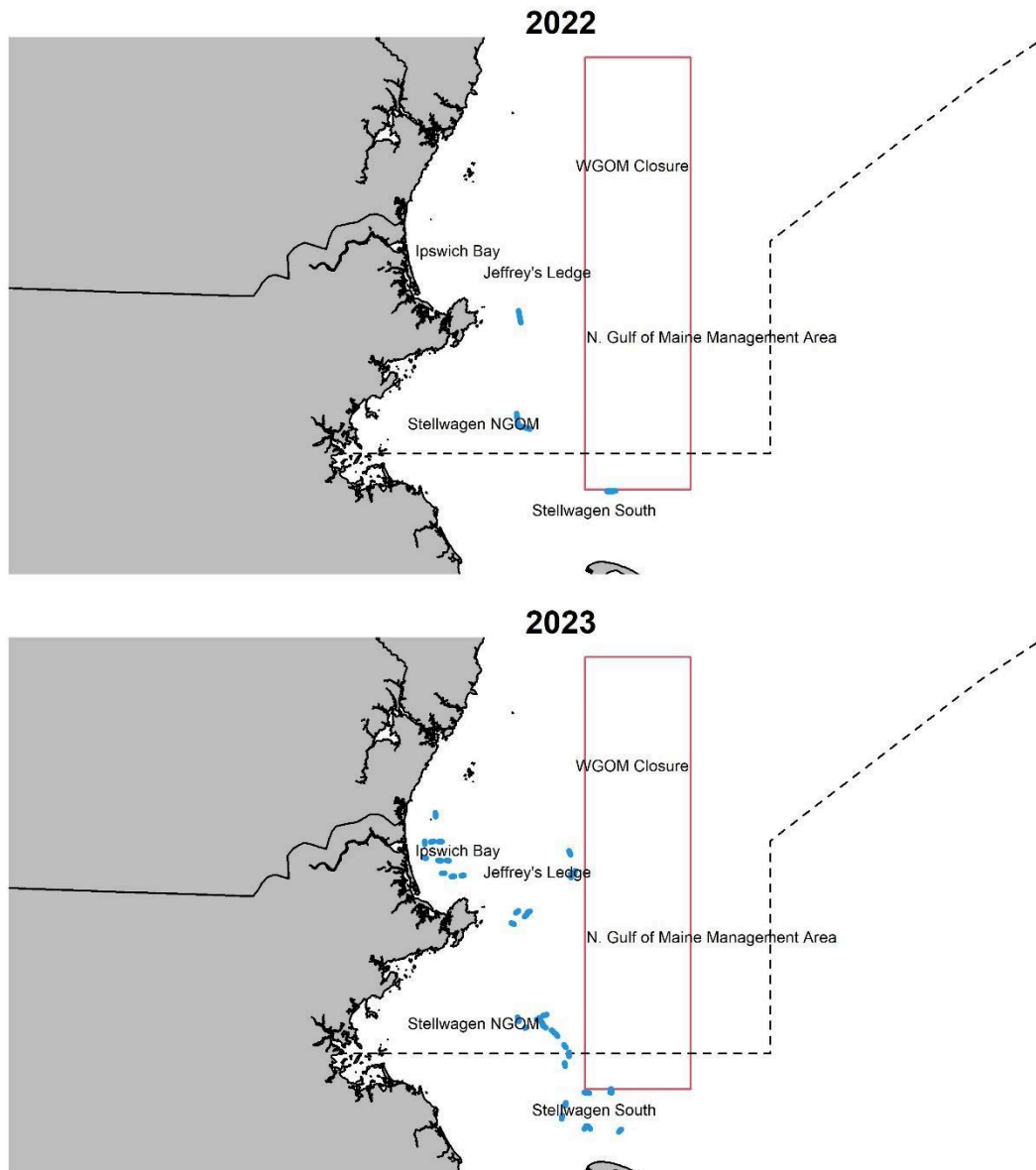


Figure 20. SMAST biological scallop dredge survey tows by year in the Gulf of Maine. The blue lines denote the tow paths, the red box surrounds the Western Gulf of Maine closure area, and the dashed black line represents the southern and eastern boundaries of the Northern Gulf of Maine management area. Tow duration was 20 minutes in 2022 and 5 minutes in 2023.

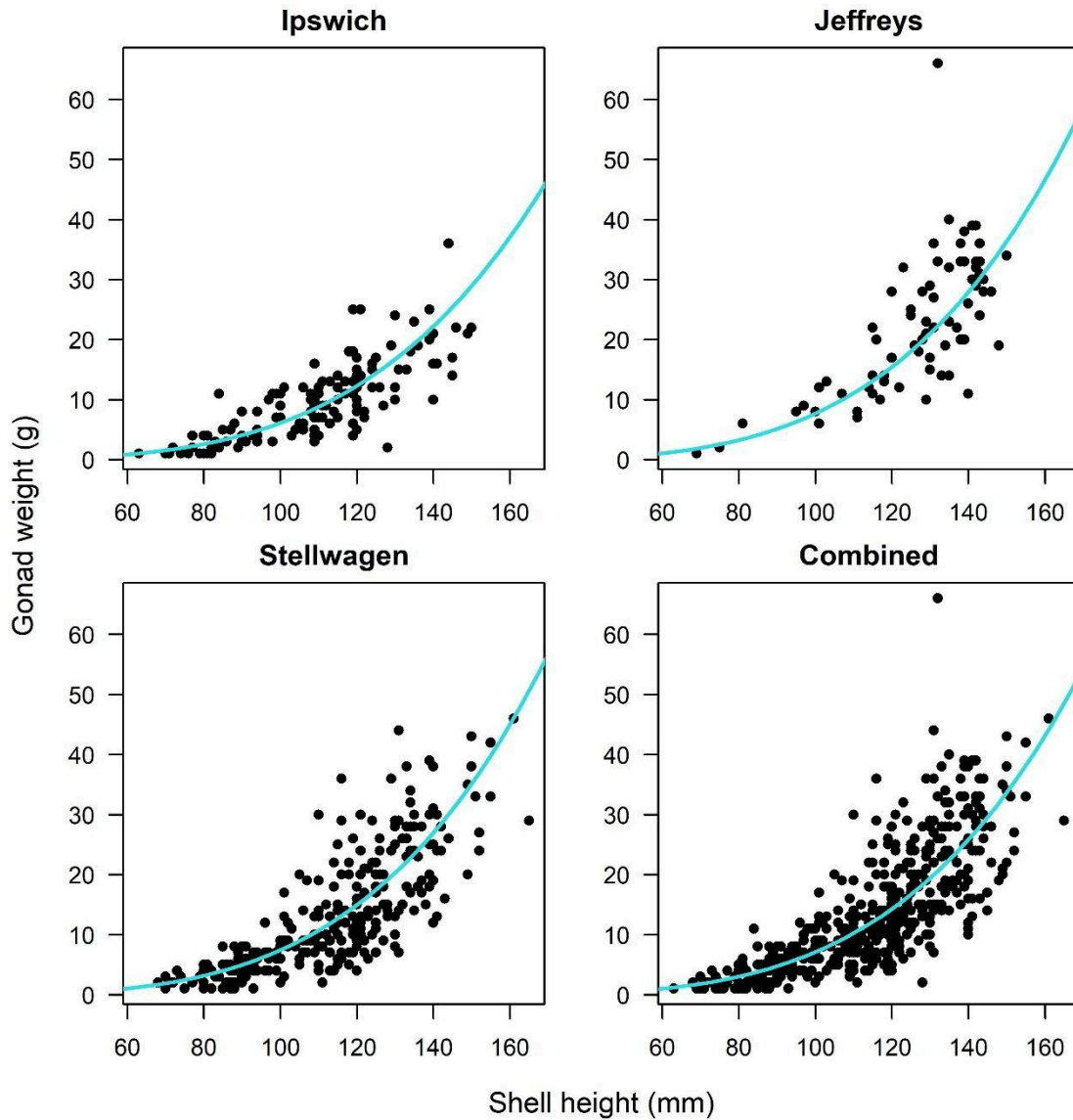


Figure 21. Gonad weight to shell height relationships derived from the SMAST 2023 dredge survey in the Gulf of Maine. Black points are the observations, each from a single scallop and the teal curves are the model fits. The combined area equation is:

$$GW = \exp \exp (-0.8985 + 3.8316 \cdot \ln \ln (SH) + 0.4666 \cdot \ln \ln (depth) - 0.3878 \cdot latitude)$$

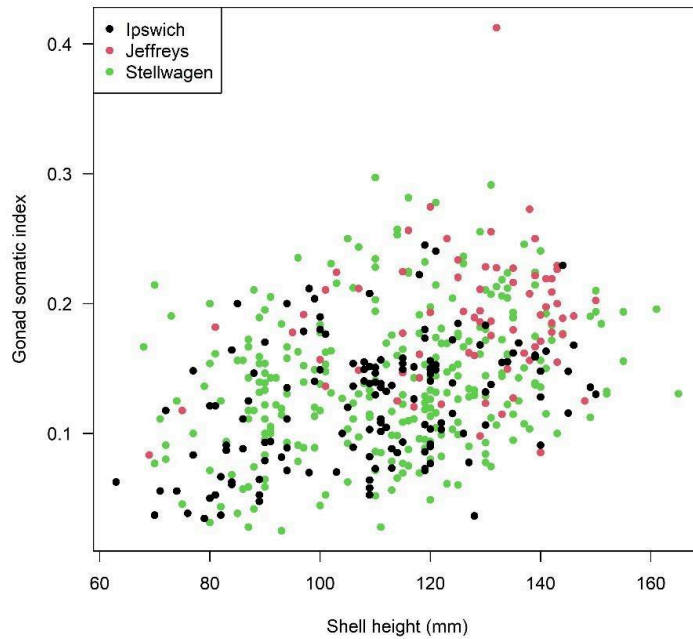


Figure 22. Gonad somatic index of scallops caught during the 2023 SMAST dredge survey in the Gulf of Maine, presented relative to shell height (mm). The observations are colored by bank. The gonad somatic index was calculated as the weight of the gonad relative to the combined weight of the abductor muscle, gonad, and all remaining soft tissue. Shell weight is not incorporated.

Table 1. Shell height (mm) to meat weight (g) equations used to convert drop camera abundance and shell height data into biomass estimates after the SMAST drop camera survey each year. In the equations W is meat weight (g), SH is shell height (mm), depth is in meters, and latitude is in decimal degrees.

Years	Name	Equation
2009 to 2017	SAW 50	$W = \exp(9.6771 + 2.8387 \cdot \ln(\text{SH}) - 0.5084 \cdot \ln(\text{depth}) - 4.7629 \cdot \ln(\text{latitude}))$
2018 to 2020	SARC/SAW 65 (GB Open)	$W = \exp(-6.69 + 2.878 \cdot \ln(\text{SH}) - 0.0073 \cdot \text{depth} - 0.073 \cdot \text{latitude})$
2021 to 2022	Hart 2020 (all other areas)	$W = \exp(-281.91 + 72.42 \cdot \ln(\text{latitude}) + -0.212 \cdot \ln(\text{depth}) + (71.13 - 18.16 \cdot \ln(\text{latitude})) \cdot \ln(\text{SH}))$
2021 to 2022	UMaine/DMR 2021 (Stellwagen areas)	$W = \exp(90.21118 + 2.894163 \cdot \ln(\text{SH}) - 26.79 \cdot \ln(\text{latitude}) - 0.07074 \cdot \ln(\text{depth}))$
2023	SMAST/DMR 2016-2023	$W = \exp(-10.36 + 3.032 \cdot \ln(\text{SH}) - 0.2155 \cdot \ln(\text{depth}) + .001 \cdot \text{latitude})$

Table 2: Annual estimates of scallop abundance (millions of scallops) and biomass (thousands of metric tons of scallop meats) from SMAST drop camera surveys of various areas within the Gulf of Maine. In addition, the area covered by the survey (km²) and the standard error (SE) of the biomass are also presented. Abundance and biomass are for all sizes of scallops greater than or equal to 40 mm in shell height. Years with “NA” for biomass and standard error indicate that no shell height data were collected to convert from shell height to biomass using a shell height meat weight equation.

Area	Year	Area Sampled (km ²)	Scallop abundance (millions)	Biomass (thousand tons)	SE of biomass
Cashes	2009	58	12.68	42.75	45.05
Cashes	2010	65	5.89	23.89	35.68
Cashes	2011	64	3.53	28.16	46.02
Cashes	2013	61	15.3	43.74	65.15
Cashes	2014	67	37.05	222.09	437.94
Cashes	2015	60	5.81	NA	NA
Cashes	2018	50.6	1.86	38.14	43.56
Cashes	2023	41.2	3.55	47.76	61.91
Fippennies	2009	81	90.37	248.07	87.64
Fippennies	2010	84	131.17	413.04	154.41
Fippennies	2011	82	142.5	634.96	263.88
Fippennies	2013	84	115.35	716.1	278.69
Fippennies	2014	85	171.91	1071.84	313.62
Fippennies	2015	84	69.82	NA	NA
Fippennies	2018	72	40.97	768.42	235.62
Fippennies	2023	68.6	32.41	562.64	173.96
Ipswich Federal	2018	94.3	10.18	216.85	112.25
Ipswich Federal	2020	81.5	4.64	70.68	24.57
Ipswich Federal	2021	79.7	7.71	110.52	46.8
Ipswich Federal	2022	79.7	7.9	95.08	38.61
Ipswich Federal	2023	79.7	7.66	68.07	18.29
Ipswich State	2018	25.7	4.81	75.85	25.88
Ipswich State	2020	22.3	1.29	33.91	18.93
Ipswich State	2021	21.4	2.57	33.16	19.01
Ipswich State	2022	22.3	1.76	19.63	7.72
Ipswich State	2023	22.3	1.75	14.57	7.08
Jeffreys	2018	65.2	4.57	23.21	13.18
Jeffreys	2020	225.5	7.97	197.52	73.67
Jeffreys	2021	93.5	11.97	207.11	60.4
Jeffreys	2022	184.4	7.22	128.86	37.16
Jeffreys	2023	155.2	7.12	115.23	29.6
Jeffreys WGOM	2009	67	38.84	164.67	82.5

Jeffreys WGOM	2010	70	33.21	244.7	176.02
Jeffreys WGOM	2011	60	38.91	369.2	223.83
Jeffreys WGOM	2013	72	43.19	341.34	144.28
Jeffreys WGOM	2014	72	27.18	326.84	131.05
Jeffreys WGOM	2015	71	26.09	NA	NA
Jeffreys WGOM	2023	190.4	27.05	378.14	108.46
Platts	2009	92	94.94	175.81	129.71
Platts	2010	91	66.31	281.36	176.14
Platts	2011	91	31.48	202.98	100.32
Platts	2013	91	67.93	181.63	92.81
Platts	2014	91	33.57	227.45	115.55
Platts	2015	91	10.39	NA	NA
Platts	2018	78	2.45	73.63	45.23
Platts	2020	78	2.57	49.35	18.18
Platts	2021	29.2	5.81	83.48	25.66
Platts	2022	77.2	4.88	93.94	41.54
Platts	2023	77.2	13.97	47.27	25.7
Stellwagen NGOM	2017	153.6	14.28	368.75	162.13
Stellwagen NGOM	2018	108	6.75	253.42	80.39
Stellwagen NGOM	2020	162.1	120.56	1805.71	1094.95
Stellwagen NGOM	2021	146.6	89.1	1208.45	656.09
Stellwagen NGOM	2022	146.6	53.97	1069.17	641.65
Stellwagen NGOM	2023	113.2	38.28	892.32	446.42
Stellwagen South	2013	710	137.03	1440.93	1026.59
Stellwagen South	2018	255.5	8.67	235	53.54
Stellwagen South	2020	255.5	42.12	861.56	194.07
Stellwagen South	2021	249.5	24.83	414.49	78.05
Stellwagen South	2022	253	23.4	254.28	51.92
Stellwagen South	2023	261.5	44.89	306.04	74.27
WGOM	2018	51.4	2.88	110.24	52.06
WGOM	2020	51.4	135.96	2949.03	664.43
WGOM	2021	51.4	84.32	1903.68	508.57
WGOM	2022	53.2	50.58	1675.64	575.8
WGOM	2023	79.7	39.58	1386.98	300.3

4. Maine DMR GOM Scallop Dredge Surveys

a. ME DMR Scallop Surveys in Federal Waters

The Maine DMR has conducted scallop dredge surveys in the federal waters of the Gulf of Maine since 2009. These surveys are conducted aboard contracted commercial scallop vessels and utilize a standardized survey dredge. The DMR survey dredge is a 7-ft wide New Bedford-style chain sweep dredge with 2-inch rings, 1¾-inch head bale, 5-inch twine top, and a 10-inch pressure plate. The dredge is equipped with

rock chains, and it is unlined. The dredge was designed to optimize sampling in the rocky substrates that are commonly encountered in the Gulf of Maine and has been the standard for dredge surveys conducted in this area to date.

This survey is a density-stratified random grid survey. The survey grid was created by overlaying a 1-km grid across the Gulf of Maine. Each grid cell was stratified into three expected scallop densities (high, medium, or low), based on past survey data, VTR records, and fleet interviews. These survey strata have been updated when new data became available as the survey progressed. The survey domain, subareas and allocations are based on management priorities and have changed over time (Figure 23).

Prior to each survey the sampling grids were randomly selected. The survey captain has full discretion to decide the location and direction of each straight-line survey tow in the selected grid cells. The standard tow was 7 minutes in 2009 and was changed to 5 minutes in the following years. At each tow site, depth and speed were recorded, as well as the time and position at the start and end of each tow. The current protocol is to continuously record GPS data and to utilize dredge mounted sensors in order to improve accuracy of the recorded tow distance.

Although the catch sampling protocol changed slightly over the years, particularly for bycatch species, the sampling protocol for scallops remained relatively consistent. In each survey, all scallops were counted, and then scallops were measured to the nearest 0.1mm. The number of scallops that were measured was either the entire catch, a subsample of at least 100 scallops, or 10% of the catch. In early years, scallop samples were collected for shell height to meat weight relationship at every fourth tow, however, since 2019, eight samples from each tow were selected for meat weight measurements.

Biomass of each survey area is estimated using the swept area method, assuming dredge efficiency of 0.4 (Table 3). The appropriate shell height/weight relationships recommended by the PDT are used to estimate meat weights (Table 4). The overall average abundance of swept area can be estimated as: $\bar{X} = \sum_{h=1}^H W_h \bar{X}_h$ where $\bar{X}_h$ is the average abundance of swept area for stratum h , H is the total number of strata, and W_h is proportion of the area of stratum h with respect to the survey area.

These surveys show the growth of various size cohorts across the survey domain (Figure 24), the relative distribution of year classes (Figure 25), and the overall distribution of scallops throughout the region (Figure 26).

Table 3. Biomass (metric tons) and abundance (millions of scallops) estimates for each survey area from the Maine DMR federal Gulf of Maine dredge surveys. Standard error is reported in parentheses for each value. The domains used to calculate these estimates were the subareas from the 2024 survey and were slightly different then used for the season setting each year to allow improved comparison between surveys for . Biomass estimates were calculated using the shell height to meat weight relationship recommended by the PDT for the 2024 survey. Abundance estimates were not calculated for the exploratory areas sampled in 2024.

	2009		2012		2016		2019		2021		2024	
	MT	# (Mil)	MT	# (Mil)	MT	# (Mil)	MT	# (Mil)	MT	# (Mil)	MT	# (Mil)
Northern Stellwagen	38.5 (8.6)	1.3 (0.3)	28.2 (10.4)	0.8 (0.2)	1255. 7 (606.8)	68.3 (34.2)	910.9 (255)	66.9 (45.7)	1621. 1 (420.9)	101.6 (26.7)	1448.2 (774.9)	39.7 (19.9)
Southern Stellwagen							611.8 (152)	23.5 (6.7)	330.4 (104.)	16.8 (7)	741.1 (357.9)	37 (18.4)
Jeffreys Ledge	1.6 (0.96)	0 (0)	6.5 (2.5)	0.3 (0.1)	239.2 (145.8)	14.4 (9.2)	1152.8 (800.3)	34 (24)			586.8 (398.1)	18.6 (13.2)
Ipswich Bay	18.7 (3.2)	0.6 (0.1)	40.5 (6.9)	2 (0.4)	110.5 (29.2)	4.4 (1.2)	242.2 (74.7)	7.7 (2.3)			474.3 (212.9)	18.6 (7.7)
Platts Bank	179.5 (117)	213.3 (144.9)	177.7 (72.7)	24.2 (10.3)	177.5 (108.7)	10 (6.1)	104.0 (81.6)	2.8 (2.2)			24.3 (15.7)	1.8 (1.3)
Fippennies Ledge	1455. 8 (536)	366.3 (138.7)	2421.2 (598.9)	217.9 (55.5)								
Mount Desert Island	142.9 (106.2)	5.8 (3.2)	75.3 (51.9)	2.7 (1.6)	68.6 (56.9)	2.1 (1.7)						
Machias Seal Island	94.1 (32.6)	3.4 (1)	101.1 (36.7)	4.2 (1.6)	323.5 (69.6)	22.5 (4.7)	687 (217.7)	27.1 (6.9)			278.3 (106.1)	15.9 (6.1)
Exploratory Areas											*	*

Table 4. Shell height (mm) to meat weight (g) equations recommended by the scallop PDT used to calculate predicted meat weight for biomass estimates from the shell height data. In the equations: W is meat weight (g), SH is shell height (mm), depth is meters, and latitude is decimal degrees.

Years	Name	Equation
2009 to 2017	SAW 50	$W = \exp(9.6771 + 2.8387 \cdot \ln(\text{SH}) - 0.5084 \cdot \ln(\text{depth}) - 4.7629 \cdot \ln(\text{latitude}))$
2018 to 2020	SARC/SAW 65 (GB Open)	$W = \exp(-6.69 + 2.878 \cdot \ln(\text{SH}) - 0.0073 \cdot \text{depth} - 0.073 \cdot \text{latitude})$
2021 to 2022	Hart 2020 (all other areas)	$W = \exp(-281.91 + 72.42 \cdot \ln(\text{latitude}) + -0.212 \cdot \ln(\text{depth}) + (71.13 - 18.16 \cdot \ln(\text{latitude})) \cdot \ln(\text{SH}))$
2021 to 2022	UMaine/DMR 2021 (Stellwagen areas)	$W = \exp(90.21118 + 2.894163 \cdot \ln(\text{SH}) - 26.79 \cdot \ln(\text{latitude}) - 0.07074 \cdot \ln(\text{depth}))$
2023	SMAST/DMR 2016-2023	$W = \exp(-10.36 + 3.032 \cdot \ln(\text{SH}) - 0.2155 \cdot \ln(\text{depth}) + 0.001 \cdot \text{latitude})$
2024 (Stellwagen, Jeffreys, Ipswich Bay)	SMAST/DMR 2016-2024	$W = \exp(-10.09 + 3.0 \cdot \ln(\text{SH}) - 0.239 \cdot \ln(\text{depth}) + 0.00156 \cdot \text{latitude} + \text{area})$ Stellwagen= 0.082, Ipswich=-0.095 Jeffreys = 0.007
2024 (MSI & Platts Bank)	DMR 2016-2024	$W = \exp(1.284 + 2.94 \cdot \ln(\text{SH}) - 0.2098 \cdot \ln(\text{depth}) - 0.2655 \cdot \text{latitude} + \text{area})$ MSI= 0.0668, Platts Bank=-0.026

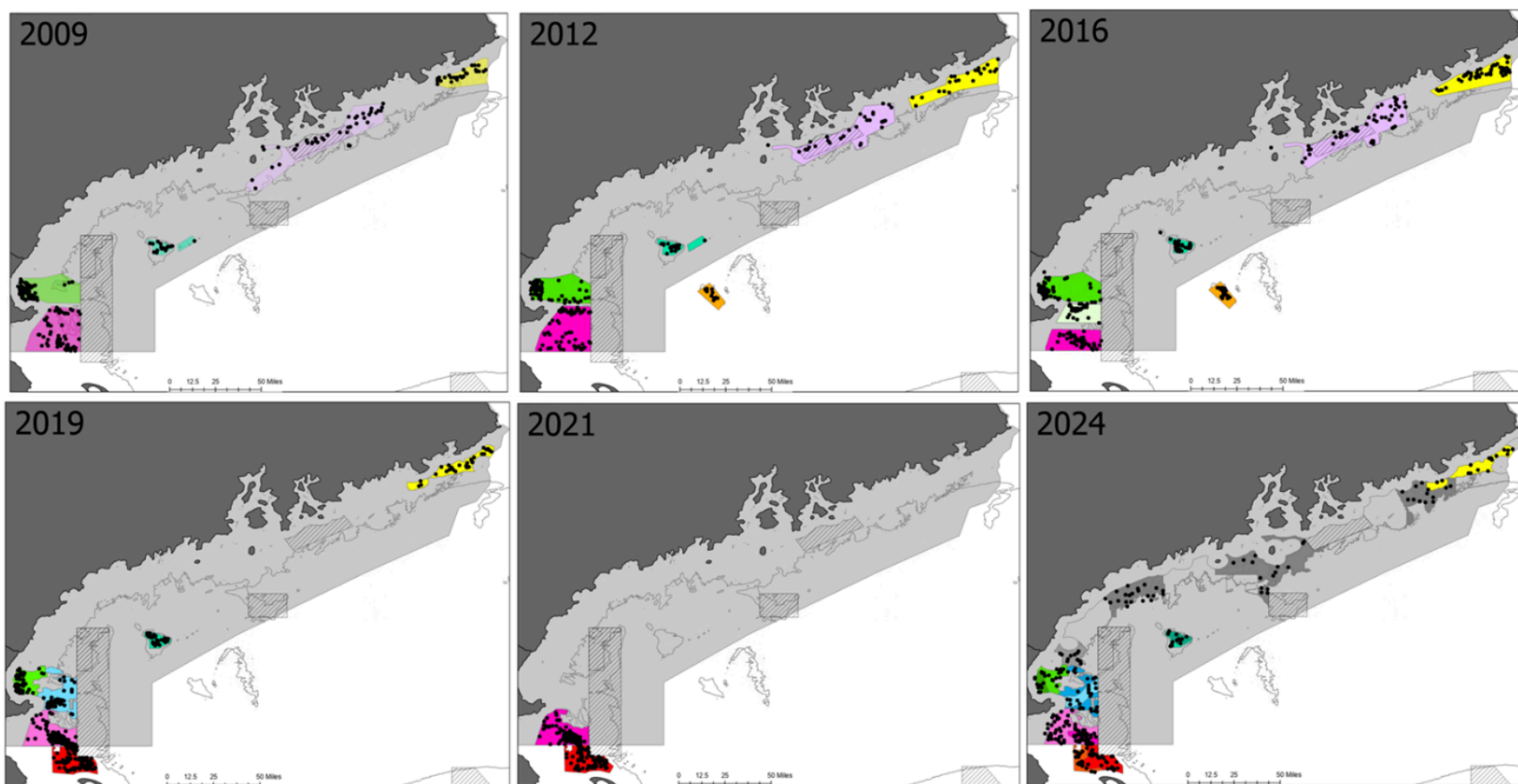


Figure 23. Maine DMR scallop dredge survey domain. The colored areas denote the survey subareas for each given year and the black points are tow locations. The NGOM area is shaded light grey. Current habitat closure areas are hashed boxes. Yellow=Machias Seal Island, Purple=Mount Dessert Island, Teal=Platts Bank, Orange=Fippennies Ledge, Green=Ipswich Bay, Blue=Jefferys Ledge, Pink=Northern Stellwagen Bank, Red=Southern Stellwagen Bank, Dark Grey=Exploratory Areas.

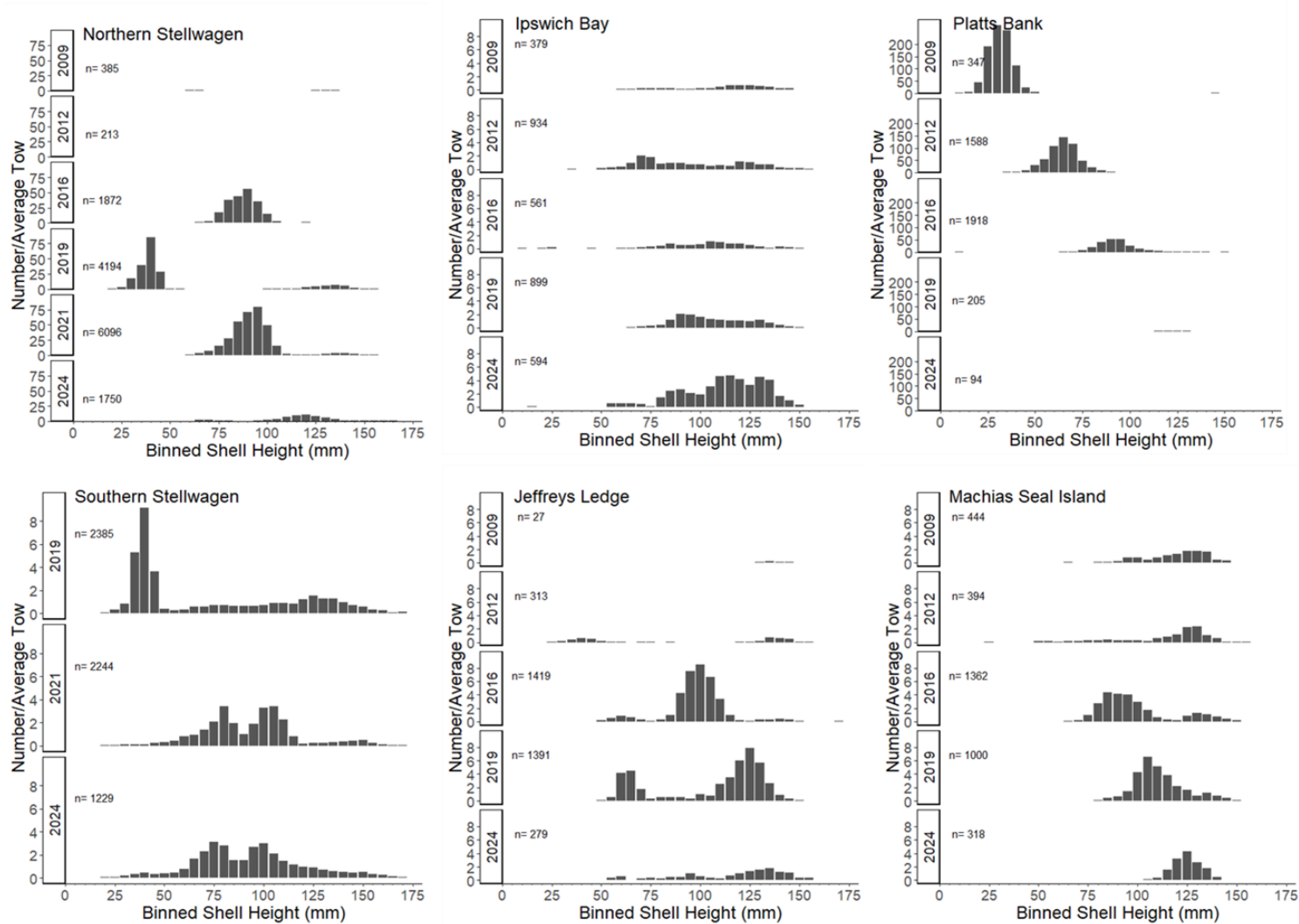


Figure 24. Binned size frequency of scallops (5 mm) for each ME DMR federal dredge survey in the Gulf of Maine. The data were grouped by the subareas from the 2024 survey domain. The number measured for each subarea is displayed for each year as text.

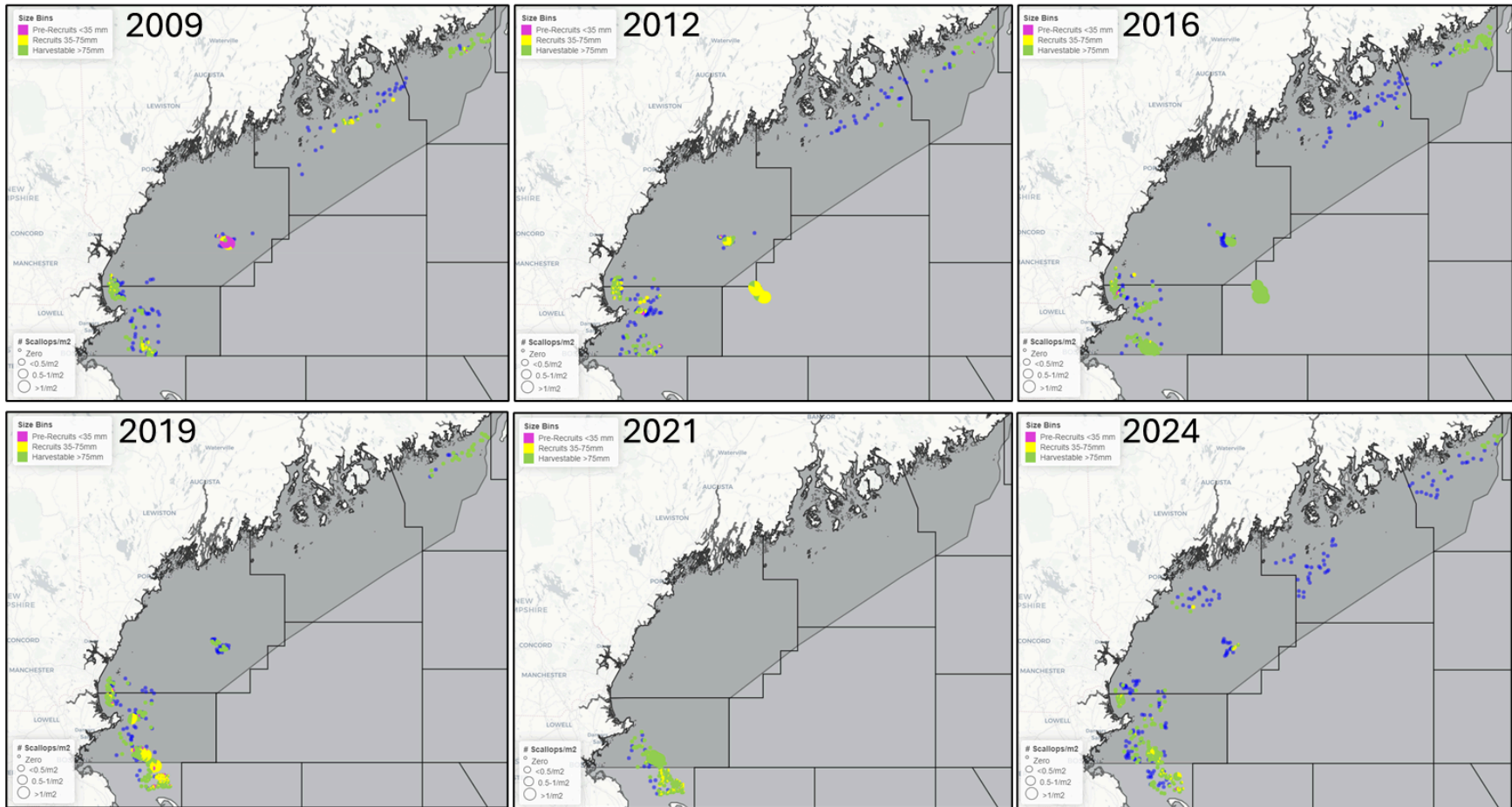


Figure 25. Scallop catch from each survey. Small circles are tows with <0.5 scallops/m², medium circles are tows with 0.5-1 scallops/m², and large circles are tows with >1 scallops/m². The color represents the proportion of each size bin (pink <35 mm, yellow 35-75 mm green >75 mm). Blue are tows with no scallops caught.

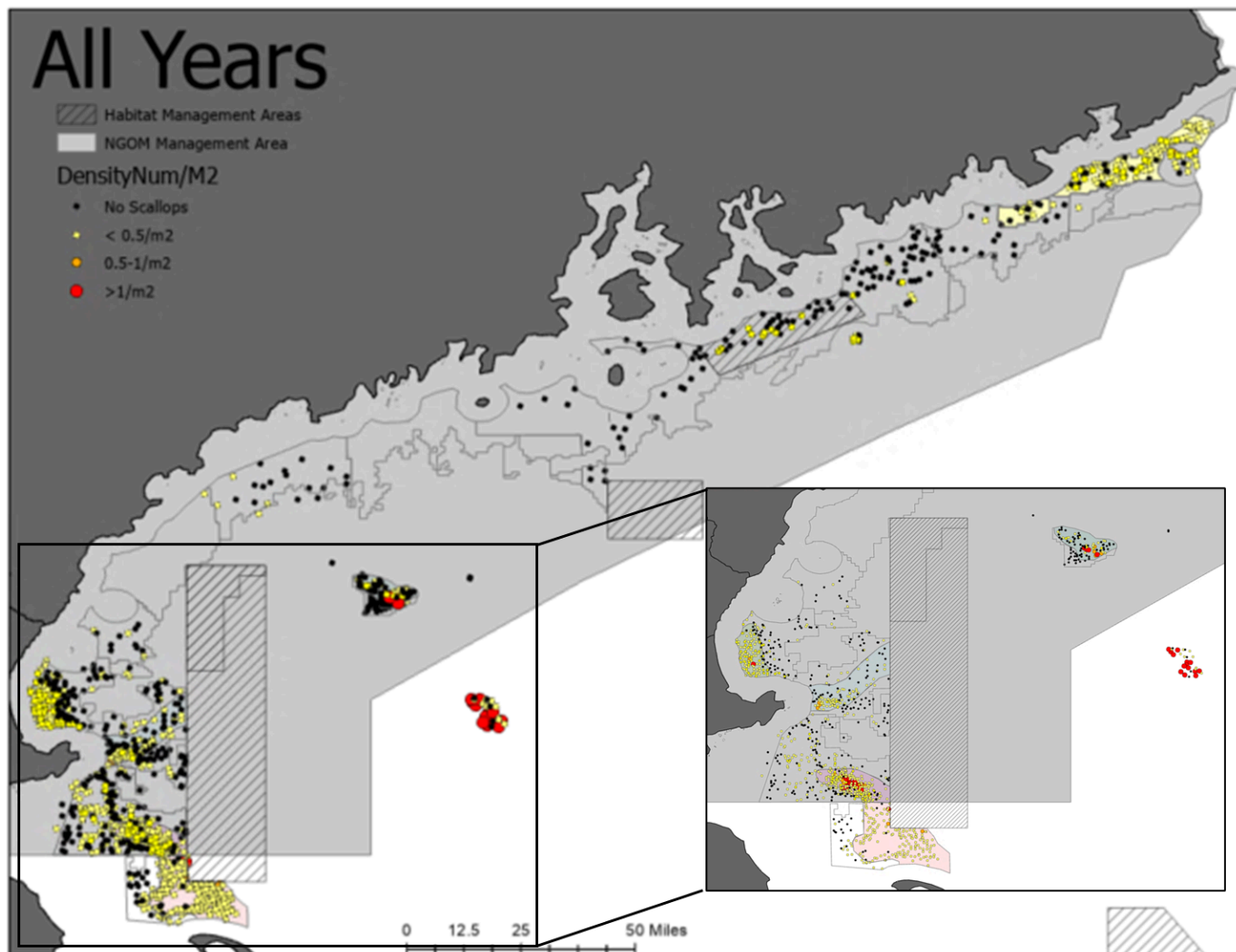


Figure 26. Scallop catch for all years (2009-2024) of the ME DMR scallop survey in federal waters. Black circles are tows with no scallops, yellow circles are tows with <0.5 scallops/m², orange circles are tows with 0.5-1 scallops/m², and red circles are tows with >1 scallops/m². The thin black lines represent the 2024 subareas used for the survey domain, and the shaded regions are the areas of interest recommended for analysis by the scallop PDT in 2024.

b. Scallop Surveys in Maine State Waters

The Maine DMR has conducted scallop resource dredge-based surveys in Maine State waters since 2005. Beginning in 2016 these surveys were modified and standardized. The surveys follow nearly the same protocols as the federal survey, however, in state waters, the target tow time is 2.5 minutes and the survey grid does not have medium density stations. The state waters are divided into three management zones (Figure 27). Zone 1 has no time series for survey data. Zone 2 has operated under a fixed rotation management plan since 2014, where 7 of the 21 subareas are open to scallop dredging for a given season. This zone is surveyed every spring, which involves sampling approximately 250 stations (~10%) in the subareas scheduled to be open in the coming season. In Zone 3, approximately 90 stations (~40%) are surveyed every fall.

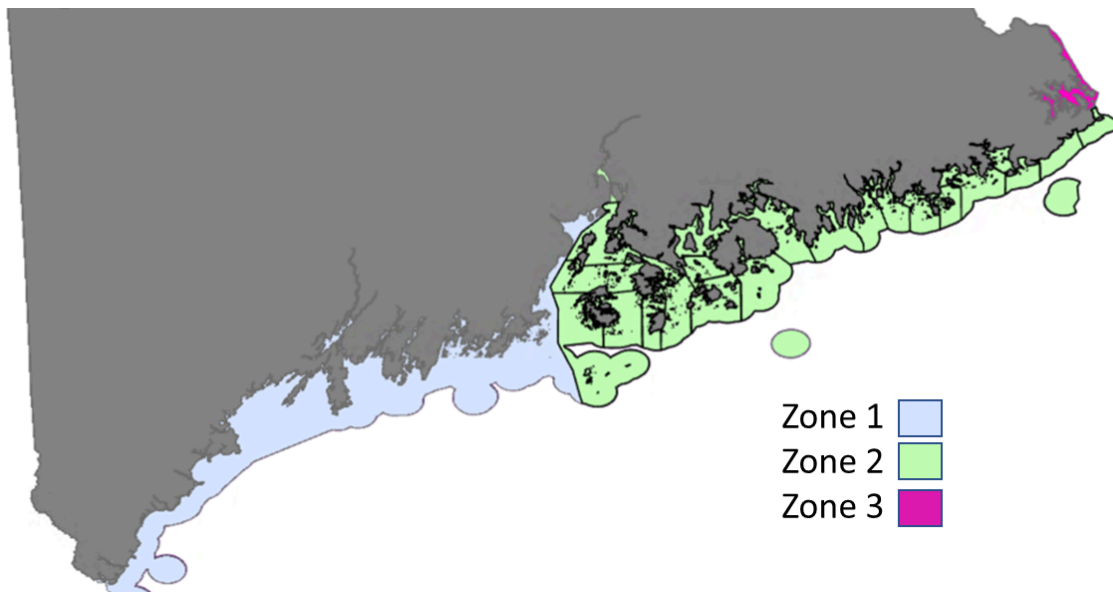


Figure 27. State of Maine Scallop management zones. The solid lines in Zone 2 are the boundaries of the 21 rotational subareas.

In Zone 2, the average catch of legal-sized (>4 inches) scallops peaked in 2018-2020 during the end of the second cycle (one cycle is 3 years) of the rotational management plan (Figure 28). Following this peak, the average legal catch steadily declined, with lower-than-average levels of sublegal scallops caught since 2022 (Figure 28). In Zone 3, the average catch of legal scallops peaked in 2021, followed by a steady decline. This peak occurred two years after the highest level of seed (<3 inches) was observed. The most recent two years saw an increase in the density of seed and sublegal scallops from the low in 2022 (Figure 29).

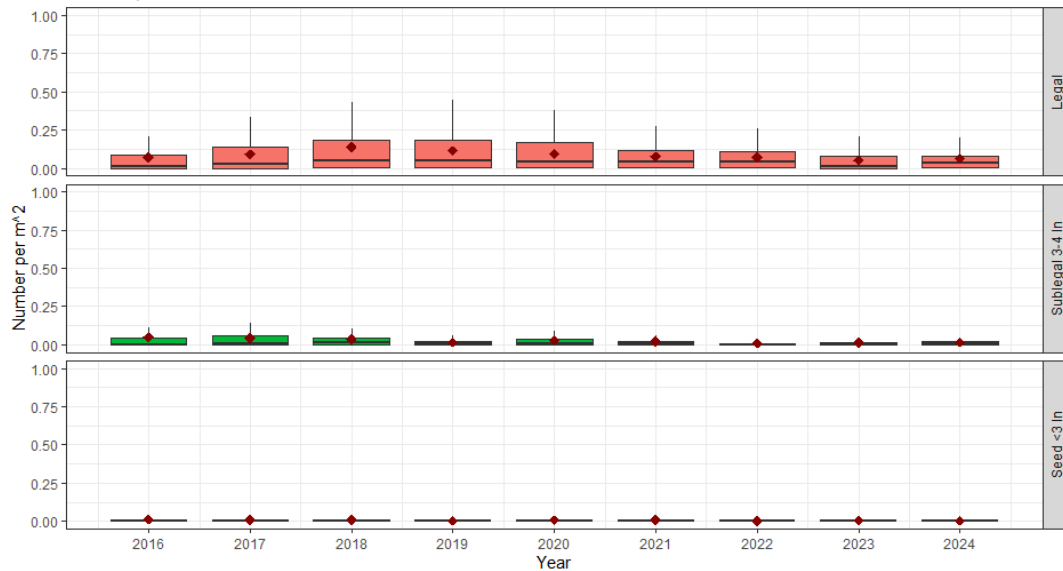


Figure 28. Maine scallop Zone 2. Average density of scallops per square meter for legal (>4 inches), sublegal (3-4 inches), and seed (<3 inches) scallops caught during the survey from 2016 until 2024 in Zone 2. This survey occurs every spring in each rotational area that is planned to be open in the upcoming season.

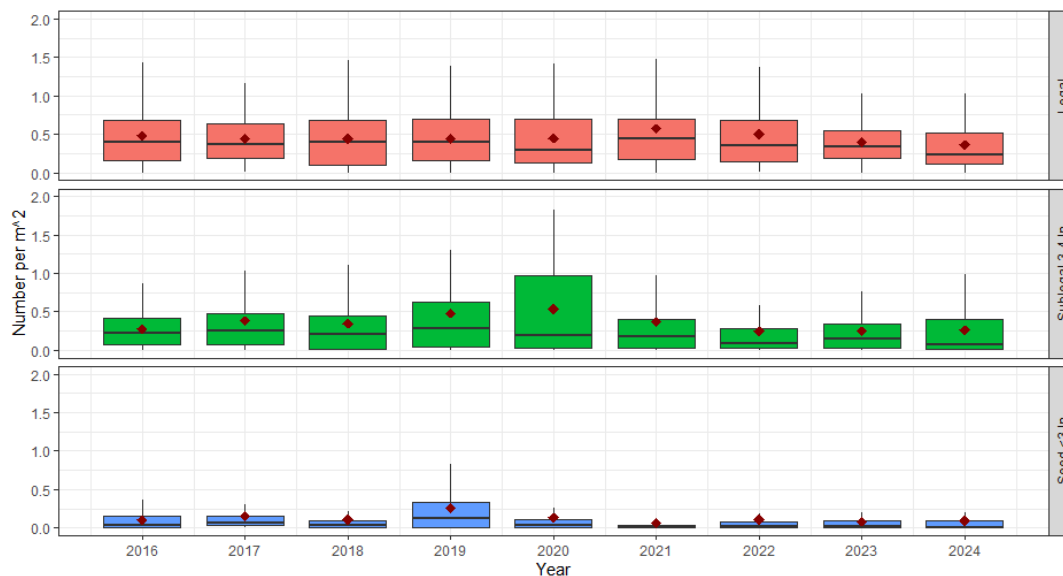


Figure 29. Maine scallop Zone 3. Average density of scallops per square meter for legal (>4 inches), sublegal (3-4 inches), and seed (<3 inches) scallops caught during the survey from 2016 until 2024 in Zone 3. This survey occurs each fall prior to the start of the season.

5. ME/NH Trawl Survey

The ME/NH Trawl survey is a biannual fishery-independent survey in the coastal waters of Maine and New Hampshire out to approximately 12 miles offshore. This survey started in the fall of 2000. The survey utilizes a modified shrimp net with a 2-inch mesh wings and 1-inch mesh liner in the cod end with an average door spread of 28.5 meters and a wing spread of 10.2 meters. The target tow duration is 20 minutes at a speed of 2.2 knots covering approximately 0.8 nm. The total scallop catch was standardized for tow length and plotted by year and mapped in 5 year bins (Figure 30).

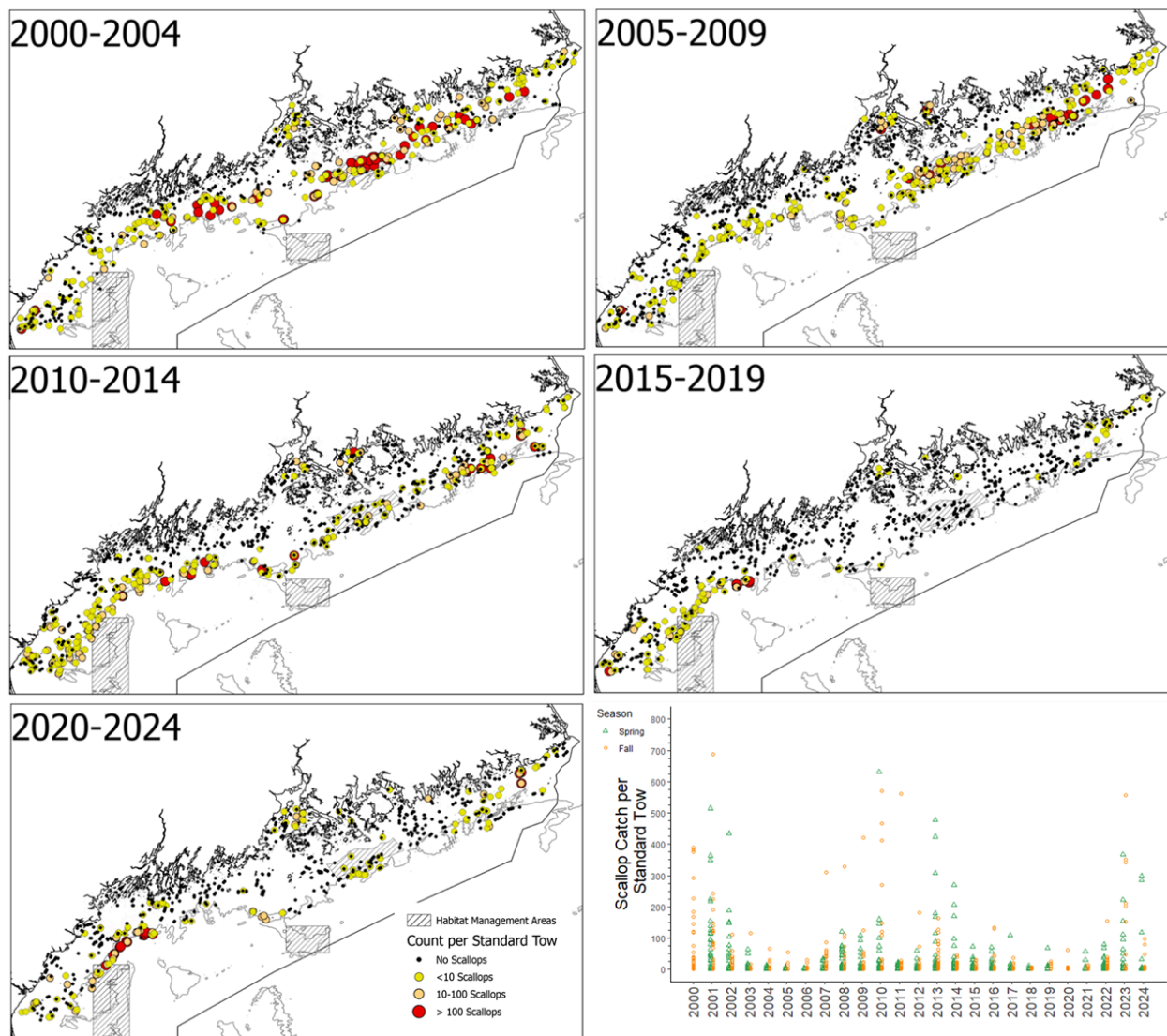


Figure 30. Maps of the scallop catch from the spring and fall ME/NH trawl survey binned into 5 year bins. Bottom left shows the total scallop catch for each tow by year for the spring (triangles) and fall (circles) ME/NH Trawl survey.

6. Methods for Developing Catch Advice

Table 5 describes the ways catch advice has been developed for the NGOM since 2008.

Starting in 2008, a total allowable landing of 70,000 pounds was set for the management area to facilitate harvest of the scallop resource in the Northern Gulf of Maine Management area. Harvest from federal waters in this of the Northern Gulf of Maine management unit had historically been a small proportion of total fishery removals, and there was limited fishery independent and dependent data available. This TAC was developed in Amendment 11 to the FMP, and was based on historic catches within the management unit.

Since 2018, the SAMS model (forecasting) has been used to develop catch advice for the Gulf of Maine, specifically areas inside and outside of the Northern Gulf of Maine Management Area. Adding portions of the GOM into the SAMS model requires estimating a “starting” biomass for the current year that can be projected forward. When the model is run, it is typically initialized using the most recent survey data. For example, survey results from 2022 were used to initialize the model for 2023 projections. As the SAMS model can be used to make multi-year projections, catch advice could be made for two or three years using the model, depending on the availability/ frequency of survey information.

Table 5 - Methods for developing catch advice for the Gulf of Maine / Northern Gulf of Maine from 2008 - 2025.

Method for developing catch advice for NGOM	Fishing Year(s) used
Historic Catch (TAC set at 70,000 lbs)	2008 – 2016
Simple exploitation of 2016 dredge survey estimates	2017
Forward projecting model using prior year's optical and/or survey data	2018 - 2025

Recommended Procedures for Developing Catch Advice in the Northern Gulf of Maine Management Unit

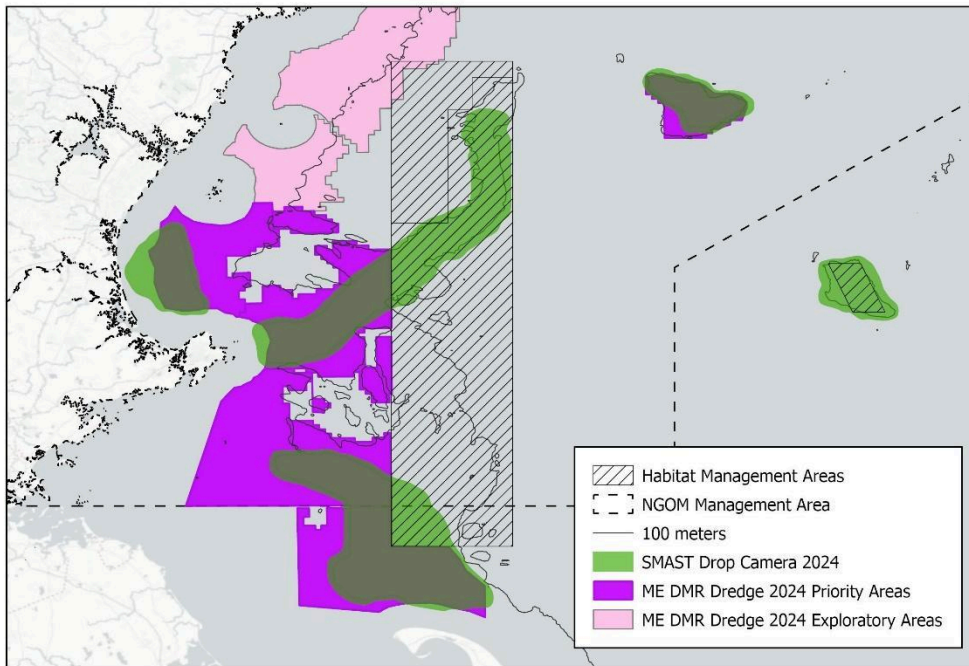
In the absence of scallop specific survey strata in the Gulf of Maine region, independent surveys have focused on areas of interest. When conducted in similar areas, the surveys often overlap, but not perfectly (Map 1, Map 2). This situation creates several decision points and considerations for the technical team assigned with developing catch advice for management, which include:

- a) Combining data from areas of survey overlap for use in management. The general practice is to average the estimates of abundance and biomass.
- b) Treatment of surveys and data that are within that same spatial area (e.g. the same bank or ledge) but do not overlap with other independent surveys.
- c) Variation in survey type and methods (e.g. optical tools, dredge surveys) that may influence where sampling can occur.

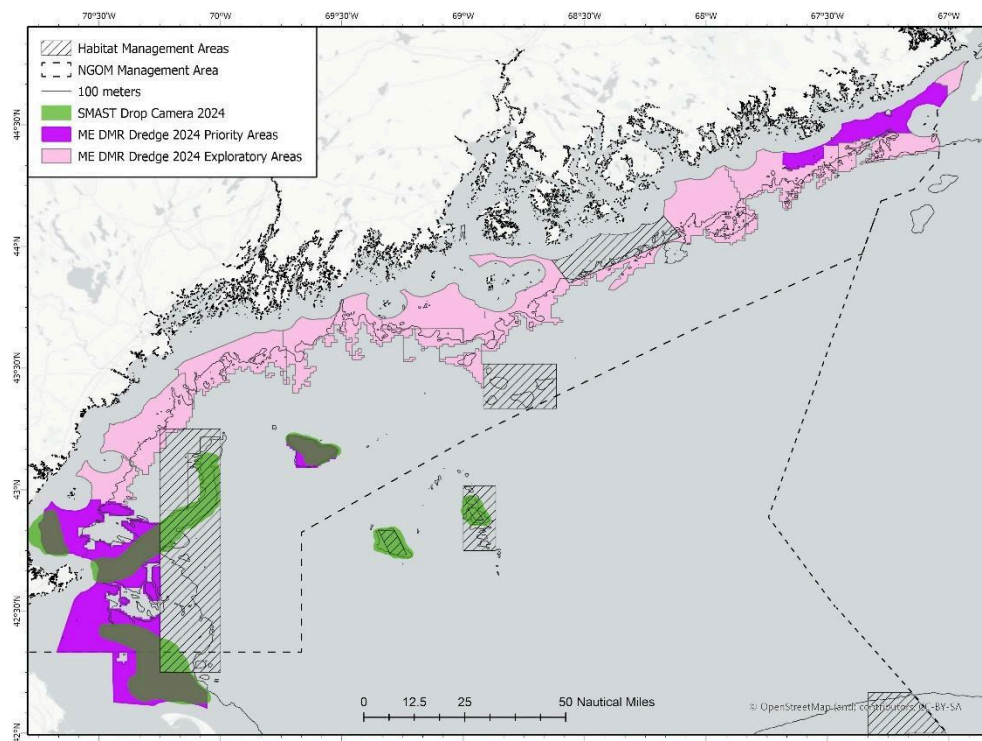
The working group recommends that steps should be taken to formalize the short-term procedure for developing catch advice:

1. Identify survey areas in the Gulf of Maine and Northern Gulf where surveys overlap, and develop “Areas of Interest” for the purposes of estimating biomass and abundance. Areas of Interest should consider both fishery independent and dependent data, including survey estimates of scallop density and fishery VMS data. In the absence of more formal survey strata, the identified Areas of Interest should be used to standardize survey coverage in the GOM and simplify the development of catch advice. These AOIs would be the basis for forward projections in the SAMS model.
2. In future years, the development of more formal survey strata should be prioritized and consider trawl survey and incidental catch data.
3. Due to the heterogenous distribution of scallops in the GOM as well as challenges sampling amidst heavy fixed-gear effort when surveys are typically conducted, multiple sampling methodologies, including Generalized Random Tessellation Stratified (GRTS), Stratified Random Sampling, and a Fixed Station optical design could be used. Catch advice should strive to be based on biomass estimates generated from both optical and dredge surveys.
4. Approaches should be developed in accordance with the Council’s Scallop Survey Working Group’s Guiding Principles in mind.²

² See: <https://www.nefmc.org/committees/scallop-survey-working-group>



Map 1 - Survey domains for inshore Gulf of Maine east of Massachusetts and New Hampshire from 2024 SMAST Drop Camera and Maine DMR Dredge Surveys.



Map 2 - Survey Domains for the entire Gulf of Maine from 2024 SMAST Drop Camera and Maine DMR Dredge Surveys.

5. Potential Methods for Developing Biological Reference Points, Data Needs

The working group considered data needs to expand the scope of the stock assessment to the Gulf of Maine. Past assessment working groups have attempted to estimate yield-per-recruit for the Gulf of Maine using the stochastic yield model (SYM) that is used to develop reference points for Georges Bank and the Mid-Atlantic. At that time, shell-height meat-weight data from UMaine/Maine DMR trawl surveys were used in the SYM model, however the model was determined to be unstable. This approach was not attempted in this research track assessment but could be tried again in future assessments.

Expanding the CASA model to the GOM region would require estimates of abundance, biomass, growth, natural mortality, incidental and discard mortality, as well as landings. As shown in earlier sections of this appendix, the survey time series has been expanded considerably since the last assessment. Additional biological and life-history information from animals in the Gulf of Maine is needed to estimate yield-per-recruit in this region, and to develop reference points for status determination

6. Overview of Federal Fishery Management – Northern Gulf of Maine

The Council established the Northern Gulf of Maine Management Area and permit category through Amendment 11. The area was developed to enable continued fishing and address concerns related to conservation, administrative burden, and enforceability of scallop fishing within the Gulf of Maine. The initial measures were intended to allow directed scallop fishing in the NGOM, and the Council envisioned that management of this area would be reconsidered if the scallop population and fishery in the NGOM grew in the future.

During development of Amendment 11 the Council did not recommend restrictions on LA vessels fishing in the NGOM because “the improved management and abundance of scallops in the major resource areas on Georges Bank and in the Mid-Atlantic region has made access to GOM scallops less important for the limited access boats and general category boats from other regions. As a result, a separate management program for scallops in the NGOM is unlikely to have any impact on these vessels” (see Amendment 11 Executive Summary NEFMC, 2007 page ix). At this time, limited access vessels were able to operate in the NGOM management area under days-at-sea (DAS) management as long as the LAGC TAC had not been caught.

Vessels receiving NGOM permits are authorized to fish within the Northern Gulf of Maine (NGOM) Management Area with a 200-pound-per-day trip limit until the annual total allowable catch (TAC) for the area is caught.

From 2009 – 2015 the NGOM TAC of 70,000 lbs was not caught, and the fishery remained open for the entire year. In FY2016 there was a notable increase in effort in the NGOM management area by both LAGC and LA vessels fishing the large year class of scallops on Stellwagen Bank. In-season monitoring estimated that the NGOM TAC was caught by the LAGC roughly 11 weeks after the start of the fishing year, and the area closed to all federally permitted vessels for the remainder of the year. Monitoring removals by the LA component in the NGOM was challenging because vessels could fish both inside and outside NGOM management area while fishing DAS on the same trip. Effort increased again in FY2017, with estimated removals topping 1.6 million pounds from a total of 105 vessels. The NGOM area closed 23 days into the fishing year once the LAGC was projected to catch their TAC.

In response to the increase in effort and landings in the NGOM area in 2016 and 2017, the Council developed a problem statement in Framework 29 for the federal scallop fishery in the NGOM management area:

Recent high landings and unknown biomass in the Northern Gulf of Maine Scallop Management Area underscore the critical need to initiate surveys and develop additional tools to better manage the area and fully understand total removals.

The scallop resource in the Gulf of Maine (including the resource in federal waters of the NGOM management area) is outside of the area considered in periodic stock assessments; however, the most recent benchmark assessment (NEFSC, 2018) did discuss the Gulf of Maine resource. The benchmark did not change how GOM scallops are considered in the overall stock assessment, but it did summarize existing data from the region and put forth recommendations regarding how catch advice might be developed in the future.

Management of the scallop fishery is aided by two key data streams: 1) annual broad scale surveys of the resource, and 2) fishery data collected by at-sea observers. These data are used to assess the condition of the scallop resource and to measure the impact of the fishery on the scallop resource, fish habitat, and non-target species caught as bycatch. While these data are available for Georges Bank and the Mid-Atlantic, the Gulf of Maine is considered a data poor region due to sporadic survey efforts and limited at-sea monitoring. The frequency of surveys in the GOM historically has followed the boom-and-bust nature of the scallop resource

Amendment 21 to the Atlantic Sea Scallop FMP restructured the management of the Northern Gulf of Maine (NGOM) by integrating it into the Acceptable Biological Catch (ABC) and Annual Catch Limit (ACL) calculations for the broader scallop fishery. Previously, NGOM allocations were treated separately and were only included in the Overfishing Limit (OFL) alongside state waters catch estimates. Under Amendment 21, NGOM biomass is factored into the ABC and ACL alongside Georges Bank and Mid-Atlantic biomass. Amendment 21 also revised how NGOM allocations are structured. Instead of a fixed hard TAC, a set-aside trigger system was implemented. When NGOM biomass exceeds 800,000 pounds, 95% of the excess allocation is directed to the NGOM APL, while 5% is set aside for LAGC IFQ vessels. Additionally, 25,000 pounds of NGOM scallops were allocated annually to the RSA program.

Following Amendment 21, Framework 34 (2022) aligned NGOM management with Amendment 21's set-aside structure, increasing the RSA by 25,000 pounds and supporting additional monitoring for directed fishing in NGOM

7. Fishery Data

The scope of fishery dependent data has expanded over time for the Northern Gulf of Maine Management Unit. More regular fishery independent surveys shifted the approach for catch setting from a TAC based on historic landings to a projection method used for developing catch advice on Georges Bank and the Mid-Atlantic. Coupled with a strong recruitment event on Stellwagen Bank, harvest limits in the management unit rose considerably after the adoption of forward projection methods (Table 6), as have the number of participants in the fishery (Figure 31).

Amendment 21 to the FMP created an observer set asides, and observer coverage (and data collection) has increased with this segment of the fishery subject to target coverage rates. Figure 32 shows the kept and discarded shell height length frequencies from observed Northern Gulf of Maine Trips in 2022 and 2023, and Figure 33 describes the tow depth and duration during 2023 Northern Gulf of Maine directed fishery.

Federally permitted scallop vessels are required to have vessel monitoring system (VMS) units to participate in the fishery. The data can be used to illustrate the intensity and spatial extent of fishing by permit type. Map 3 shows fishing activity in the Gulf of Maine in fishing year 2024. The working group notes that the majority of fishing activity occurred on Stellwagen Bank east of Boston.

In Map 3, the VMS data represent combined scallop fishery activity in terms of hours fished, aggregated at a resolution of 3 nautical mile squares with a minimum of 20 hours recorded per square. A speed filter of 2 to 5 kts was applied to remove vessel activity that was likely a result of transiting to and from fishing grounds.

Table 6 - Overview of harvest limits and season length for the Northern Gulf of Maine Management Area.

Fishing Year	TAC (lb)	LAGC TAC (lb)	TAL (lb)	Harvest Set-aside (lb)	F-rate used in projection	Season length (days)
2008-2015	70,000					365
2016	67,454					73
2017	75,000					22
2018	200,000	135,000				31
2019	205,000	137,500				24
2020	350,000	190,000				52
2021	175,000	122,500				35
2022			661,387	621,307	0.18	55
2023			434,311	380,855	0.15	20
2024			454,152	420,598	0.21	19
2025			712,093	675,563	0.18	

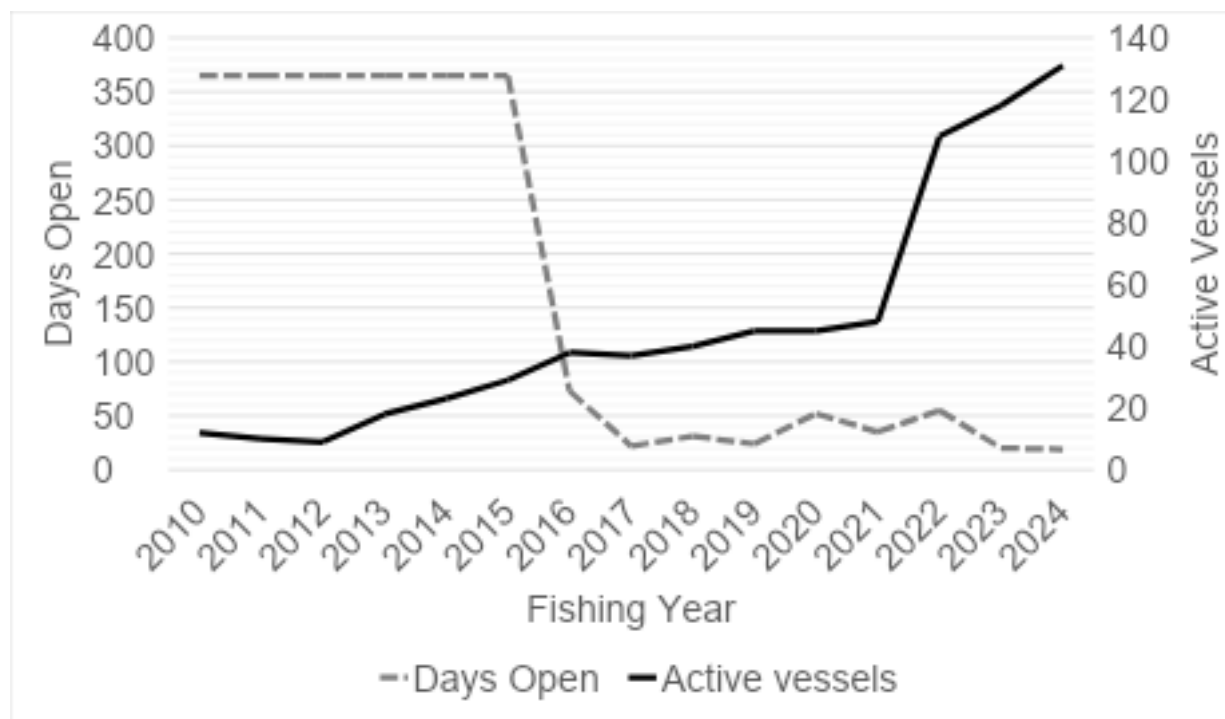


Figure 31 - Length of NGOM Season (days) and number of active vessels in the Northern Gulf of Maine for fishing years 2010 - 2024.

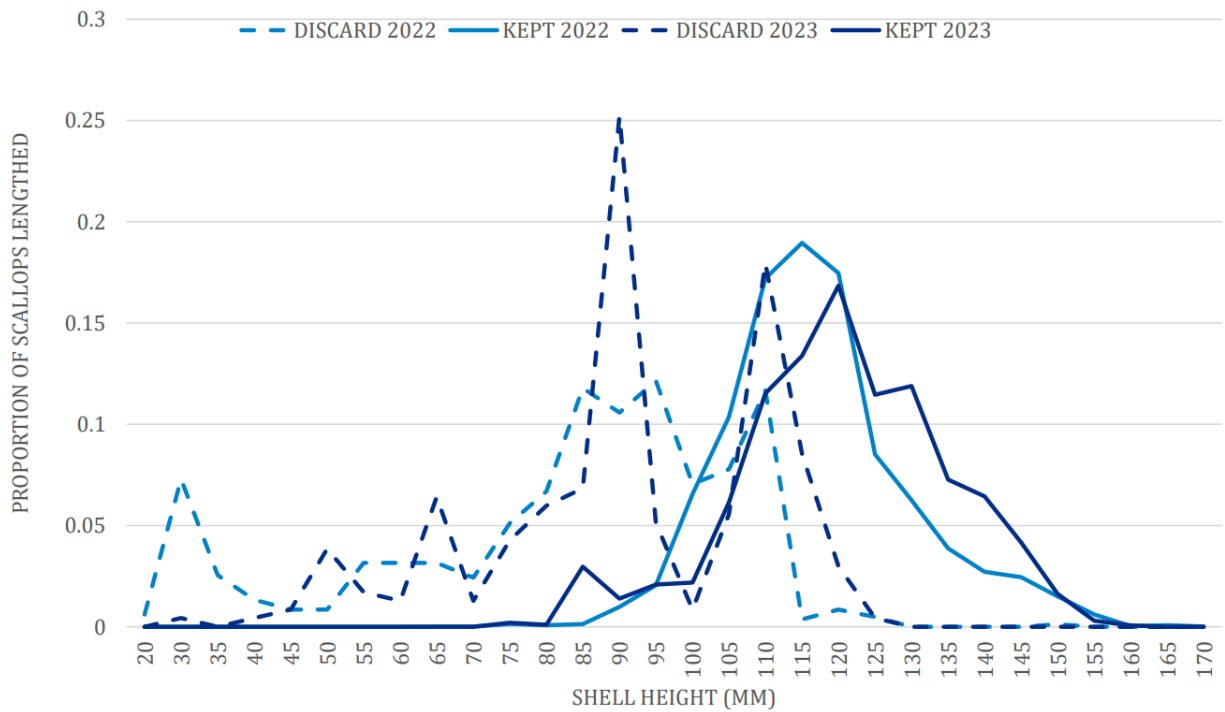


Figure 32 - Kept and Discarded Shell Height Length Frequencies from observed Northern Gulf of Maine Trips in 2022 and 2023 (Source: NEFSC Observer Program).

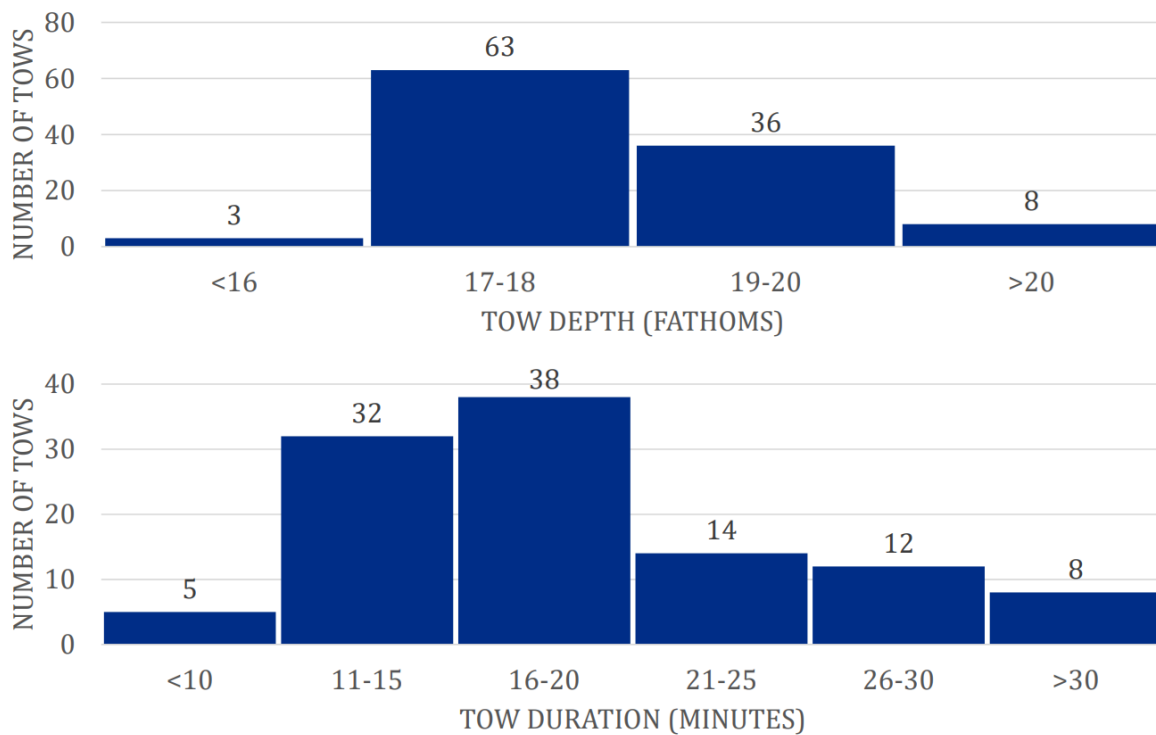
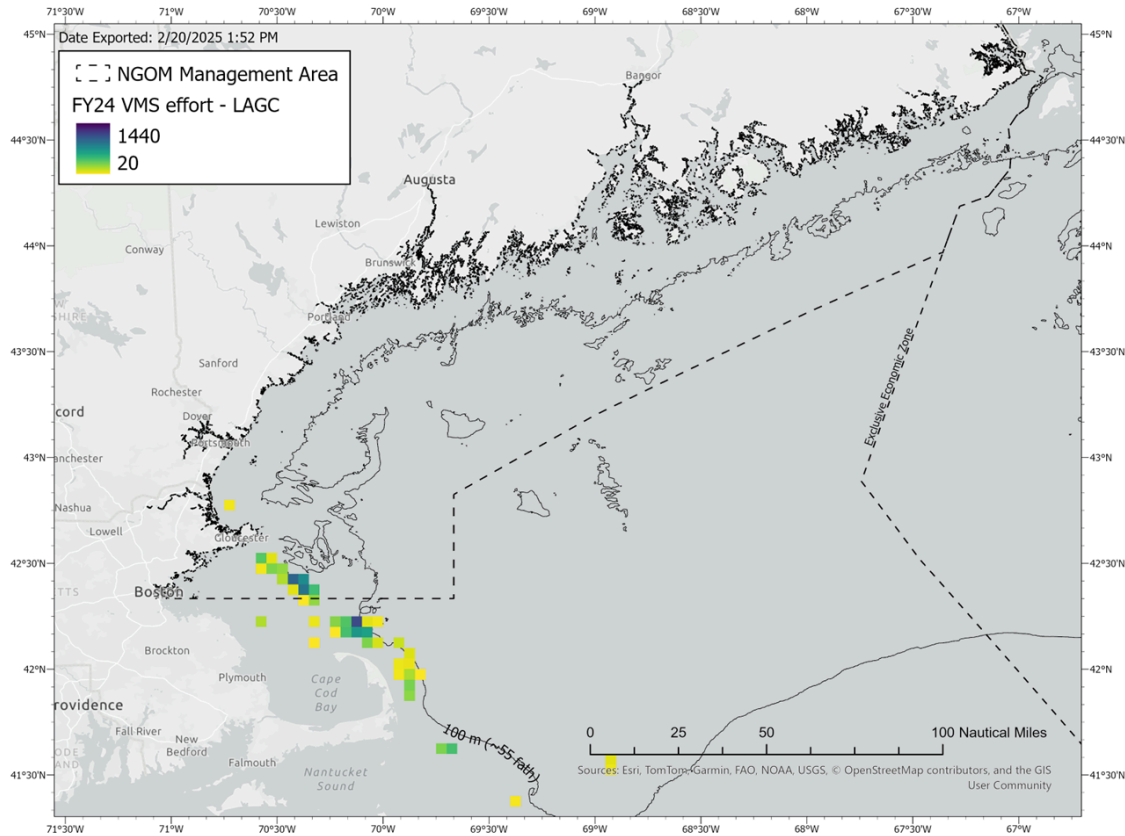


Figure 33 - Tow depth and duration during 2023 Northern Gulf of Maine directed fishery (source: NEFSC Observer Program)



Map 3 - Fishing Year 2024 Vessel Monitoring System effort (hours fished).

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